

Microeconomics (Social Choice Theory)

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Chapter 1. Preliminaries: Mathematics

0. Review of Mathematics in $\mathbb{R}^n$

Definition 0.1. Euclidean Space

$\mathbb{R}^n$ is the set of n -tuples of real numbers:

$$x = (x_1, \dots, x_n) \in \mathbb{R}^n. \quad (1)$$

A **metric** on $\mathbb{R}^n$ is a function $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$. The standard choice is **Euclidean distance**:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}. \quad (2)$$

The pair $(\mathbb{R}^n, d)$ is called **Euclidean space**.

For $x \in \mathbb{R}^n$ and $\epsilon > 0$, the **open ball** centered at x with radius ϵ is

$$B_\epsilon(x) = \{y \in \mathbb{R}^n \mid d(x, y) < \epsilon\}. \quad (3)$$

Definition 0.2. Open and Closed Sets

A set $A \subseteq \mathbb{R}^n$ is **open** if for every $x \in A$, there exists $\epsilon > 0$ such that

$$B_\epsilon(x) \subseteq A. \quad (4)$$

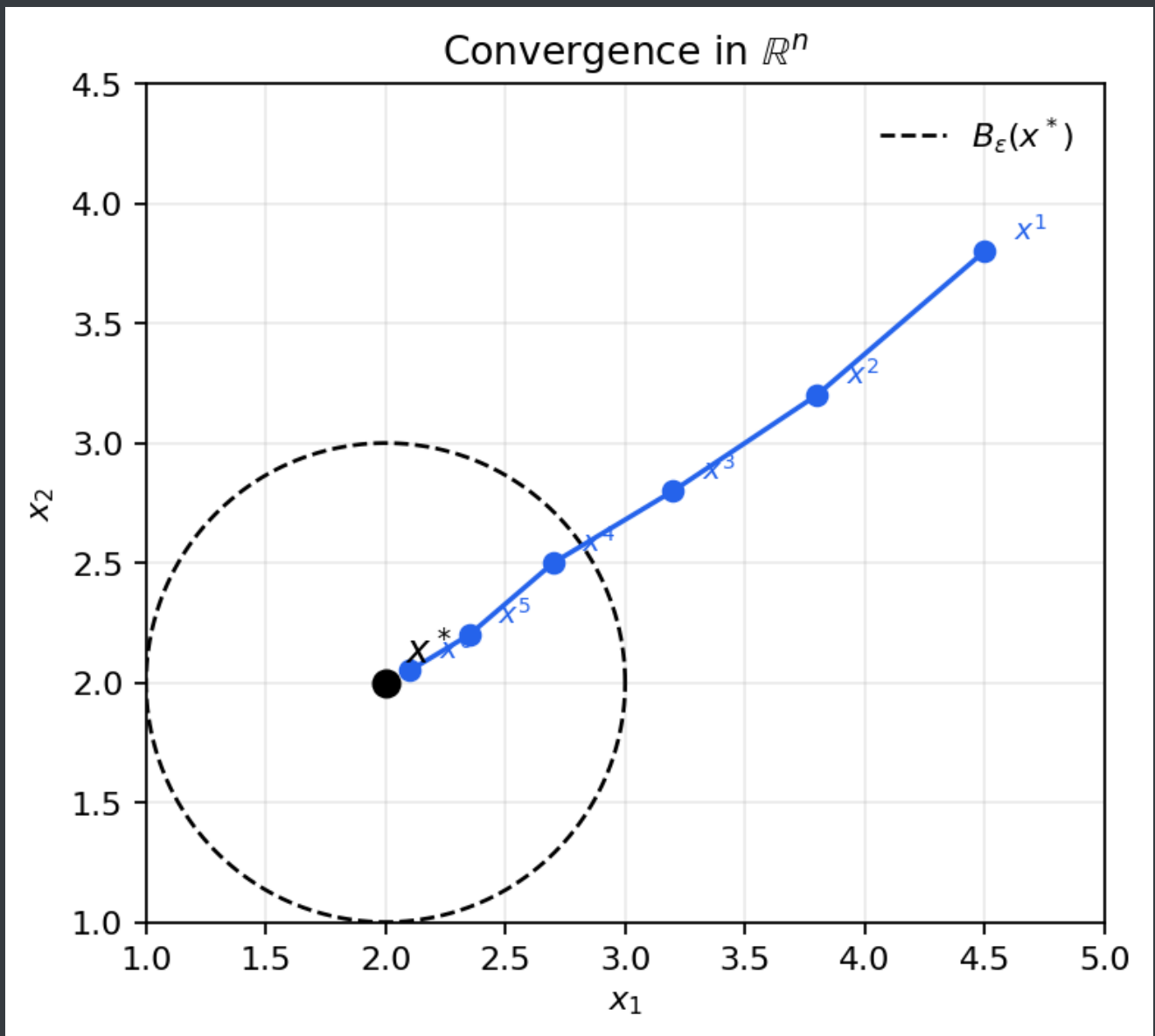
A set $A \subseteq \mathbb{R}^n$ is **closed** if its complement $A^c = \mathbb{R}^n \setminus A$ is open.

Definition 0.3. Convergence of Sequences

A sequence $\{x^n\}_{n=1}^\infty \subset \mathbb{R}^n$ **converges** to $x^* \in \mathbb{R}^n$, written $x^n \rightarrow x^*$, if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$x^n \in B_\epsilon(x^*) \quad \text{for all } n \geq N. \quad (5)$$

Figure (sequence entering an ϵ -ball).



Proposition 0.4. Sequential Characterization of Closed Sets

$A \subseteq \mathbb{R}^n$ is closed if and only if for every sequence $\{x^n\}_{n=1}^\infty \subseteq A$ with $x^n \rightarrow x^*$, we have $x^* \in A$.

Equivalently: closed sets are **sequentially closed**.

Definition 0.5. Bounded Set

$A \subseteq \mathbb{R}^n$ is **bounded** if there exists $M > 0$ such that

$$A \subseteq B_M(0). \quad (6)$$

Definition 0.6. Compact Set (in $\mathbb{R}^n$)

$A \subseteq \mathbb{R}^n$ is **compact** if A is closed and bounded.

Remark (Heine-Borel). In $\mathbb{R}^n$ with the usual topology, compact $\iff$ closed and bounded. This equivalence does **not** hold in arbitrary metric spaces.

Definition 0.7. Connected Set

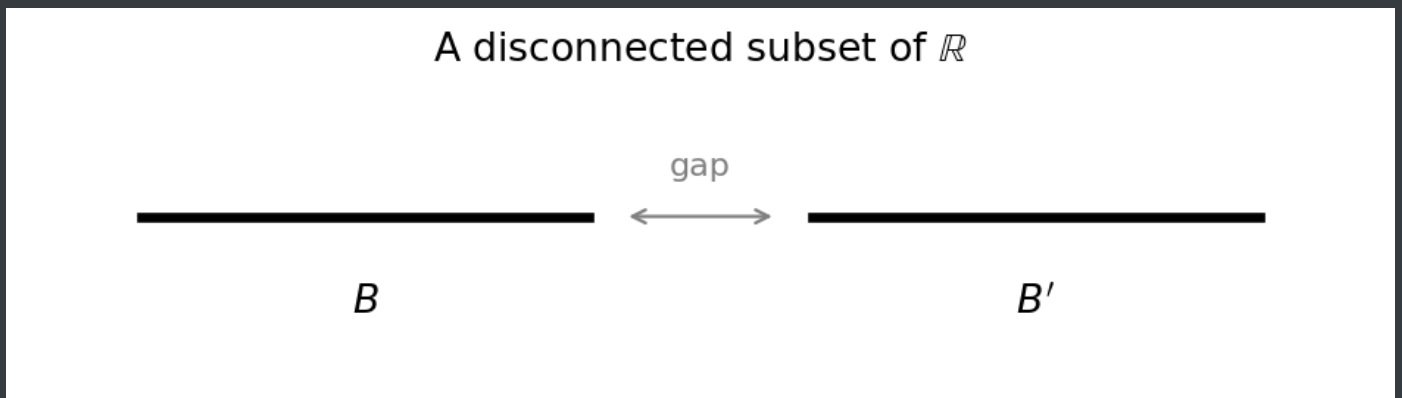
$A \subseteq \mathbb{R}^n$ is **disconnected** if there exist open sets $U, V \subseteq \mathbb{R}^n$ such that

$$A \subseteq U \cup V, \quad A \cap U \neq \emptyset, \quad A \cap V \neq \emptyset, \quad A \cap U \cap V = \emptyset. \quad (7)$$

A is **connected** if it is not disconnected.

Correction from handwritten notes. The original sketch listed open sets B, B' covering A but did not require both $A \cap B$ and $A \cap B'$ to be nonempty. Without that, every set would be trivially "disconnected."

Figure (a disconnected subset of $\mathbb{R}$).



1. Continuity of Functions

Definition 1.1. Real-Valued Function on $\mathbb{R}^n$

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ assigns to each $x \in \mathbb{R}^n$ a real number $f(x) \in \mathbb{R}$.

Definition 1.2. Continuity

f is **continuous at** $x \in \mathbb{R}^n$ if for every sequence $\{x^n\}_{n=1}^{\infty} \subset \mathbb{R}^n$ with $x^n \rightarrow x$,

$$f(x^n) \rightarrow f(x). \quad (8)$$

f is **continuous** (on $\mathbb{R}^n$) if it is continuous at every $x \in \mathbb{R}^n$.

Proposition 1.3. Equivalent Characterizations of Continuity

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the following are equivalent:

1. f is continuous.
2. For every $c \in \mathbb{R}$, the **lower contour set** $\{x \in \mathbb{R}^n \mid f(x) \leq c\}$ is closed.
3. For every $c \in \mathbb{R}$, the **upper contour set** $\{x \in \mathbb{R}^n \mid f(x) \geq c\}$ is closed.

This link between continuity and closed contour sets is what we use later when defining **continuous preferences** via closed upper/lower contour sets.

Chapter 2. Rationality: Preference and Choice

2. Preference Relations and Rationality

Definition 2.1. Choice Set and Binary Relation

Let X be the set of alternatives (consumption bundles, menus, etc.).

A **binary relation** on X is a subset

$$\succeq \subseteq X \times X. \quad (9)$$

Write $(x, y) \in \succeq$ as $x \succeq y$: " x is at least as good as y ."

Example 2.2. Three Alternatives

Let $X = \{\text{coca, orange, coffee}\}$ and abbreviate c, o, cf . Suppose

$$\succeq = \{(c, o), (c, cf)\}. \quad (10)$$

Then coca is weakly preferred to orange and to coffee. If also $(o, c) \in \succeq$, we would say $c \sim o$.

Definition 2.3. Derived Relations

For $x, y \in X$:

$$x \succ y \iff x \succeq y \text{ and } y \not\succeq x \quad (\text{strict preference}), \quad (11)$$

$$x \sim y \iff x \succeq y \text{ and } y \succeq x \quad (\text{indifference}). \quad (12)$$

Equivalently, $x \succeq y \iff (x, y) \in \succeq$.

Definition 2.4. Rational Preference

$\succeq$ is **complete** if for all $x, y \in X$, either $x \succeq y$ or $y \succeq x$ (or both).

$\succeq$ is **transitive** if for all $x, y, z \in X$,

$$x \succeq y \text{ and } y \succeq z \implies x \succeq z. \quad (13)$$

$\succeq$ is **rational** if it is both complete and transitive.

3. Utility Functions and Representation

Definition 3.1. Utility Representation

A function $u : X \rightarrow \mathbb{R}$ **represents** $\succeq$ if for all $x, y \in X$,

$$u(x) \geq u(y) \iff x \succeq y. \quad (14)$$

Immediate implication. If such a u exists, then $\succeq$ must be rational (completeness and transitivity follow from the order on $\mathbb{R}$).

Question: When does a representing utility exist?

- On a **finite** X : rationality is enough.
- On **infinite** X (e.g. $\mathbb{R}^n$): rationality alone is **not** enough (lexicographic preferences are a standard counterexample).
- On $\mathbb{R}^n$, **rationality + continuity** suffices (Debreu's theorem; Section 4).

Proposition 3.2. Utility on Finite Sets

Suppose X is finite. $\succeq$ is rational if and only if there exists $u : X \rightarrow \mathbb{R}$ representing $\succeq$.

Proof sketch. Since X is finite and $\succeq$ is complete and transitive, we can rank all elements. One representing utility is

$$u(x) = |\{y \in X \mid x \succeq y\}|, \quad (15)$$

the number of alternatives that x is weakly at least as good as. If x is top-ranked, $u(x) = |X|$; if x is bottom-ranked, $u(x) = 1$. Transitivity ensures this is consistent with $\succeq$.

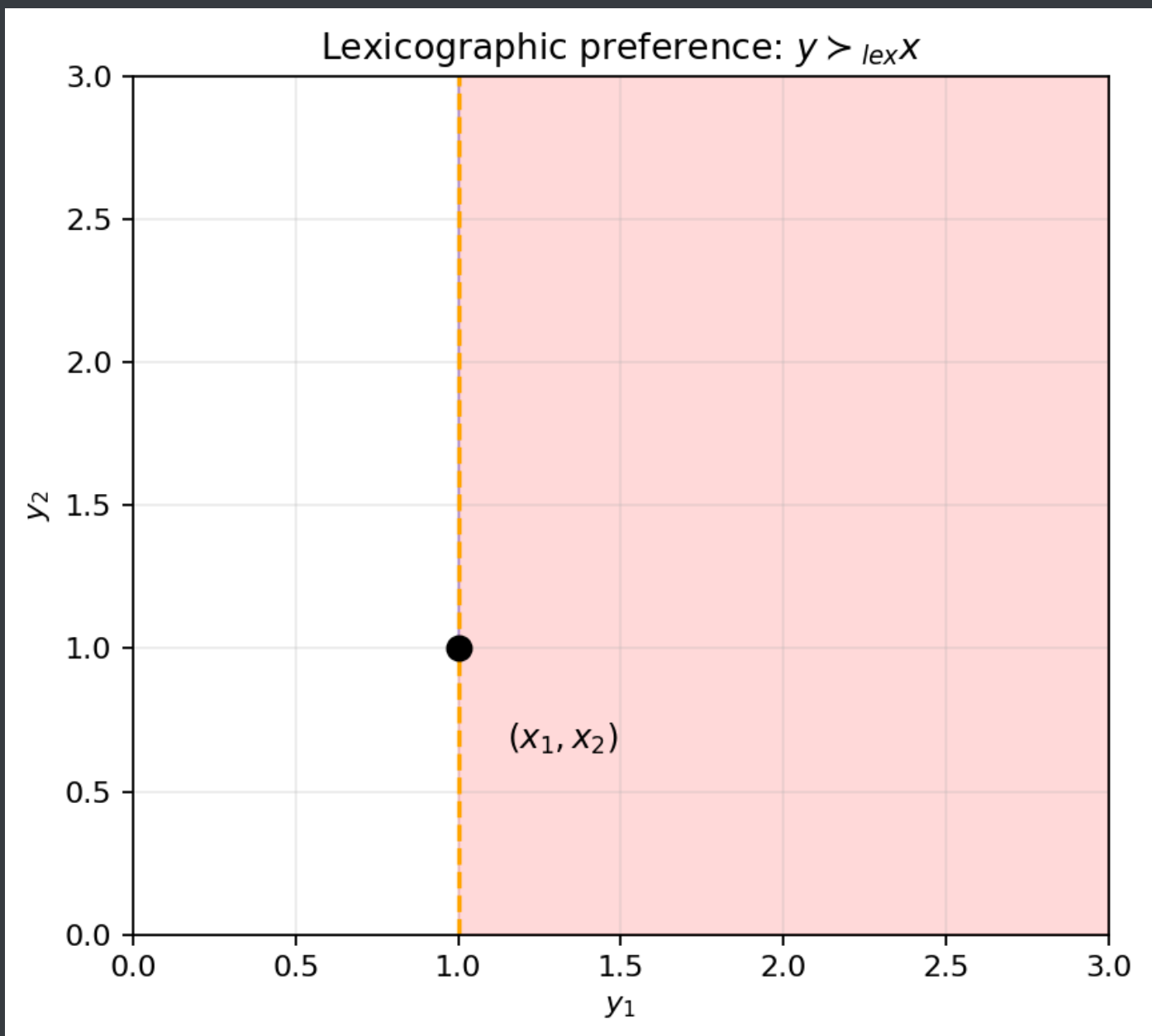
Example 3.3. Lexicographic Preferences (No Utility on $\mathbb{R}^2$)

On $X = \mathbb{R}^2$, define $\succ_{lex}$ by

$$(x_1, x_2) \succ_{lex} (y_1, y_2) \iff \begin{cases} x_1 > y_1, \text{ or} \\ x_1 = y_1 \text{ and } x_2 > y_2. \end{cases} \quad (16)$$

Define weak preference $x \succeq_{lex} y$ unless $y \succ_{lex} x$. Then $\succeq_{lex}$ is complete and transitive (hence rational), but **no continuous utility function** represents it on $\mathbb{R}^2$; in fact **no utility function at all** represents lexicographic preferences on $\mathbb{R}_+^2$ in the usual construction.

Figure (preferred bundles relative to x).



Definition 3.4. Continuous Preferences

$\succeq$ on $X \subseteq \mathbb{R}^n$ is **continuous** if for every $x \in X$, the sets

$$U(x) = \{y \in X \mid y \succeq x\} \quad (\text{upper contour}), \quad (17)$$

$$L(x) = \{y \in X \mid x \succeq y\} \quad (\text{lower contour}) \quad (18)$$

are **closed** in the subspace topology of X .

Example 3.5. Lexicographic Preferences Are Not Continuous

Let $x^n = \left(\frac{1}{n}, 0\right)$ and $y = (0, 1)$. For every n ,

$$x^n \succ_{lex} y \quad \text{because } \frac{1}{n} > 0. \quad (19)$$

But $x^n \rightarrow (0, 0)$ and at the limit

$$y \succ_{lex} (0, 0) \quad \text{because } 0 = 0 \text{ and } 1 > 0. \quad (20)$$

So the strict ranking of y versus the sequence **reverses in the limit**, which violates continuity of $\succeq_{lex}$ (equivalently, $U(y)$ or $L((0, 0))$ fails to be closed).

Proposition 3.6. Debreu Representation (Continuous Utility)

Suppose $X = \mathbb{R}^n$ (or commonly $X = \mathbb{R}_+^n$) and $\succeq$ is rational and continuous. Then there exists a **continuous** utility function $u : \mathbb{R}^n \rightarrow \mathbb{R}$ that represents $\succeq$.

Proof idea (Section 4). Under additional **monotonicity**, construct $u(x)$ as the unique α such that $x \sim \alpha \mathbf{1}_n$, where $\mathbf{1}_n = (1, \dots, 1)^\top$.

4. Continuous Preferences and Existence of Utility

We work on $X = \mathbb{R}_+^n$ (or $\mathbb{R}^n$) with a rational, continuous preference $\succeq$.

Definition 4.1. Monotonicity (used in construction)

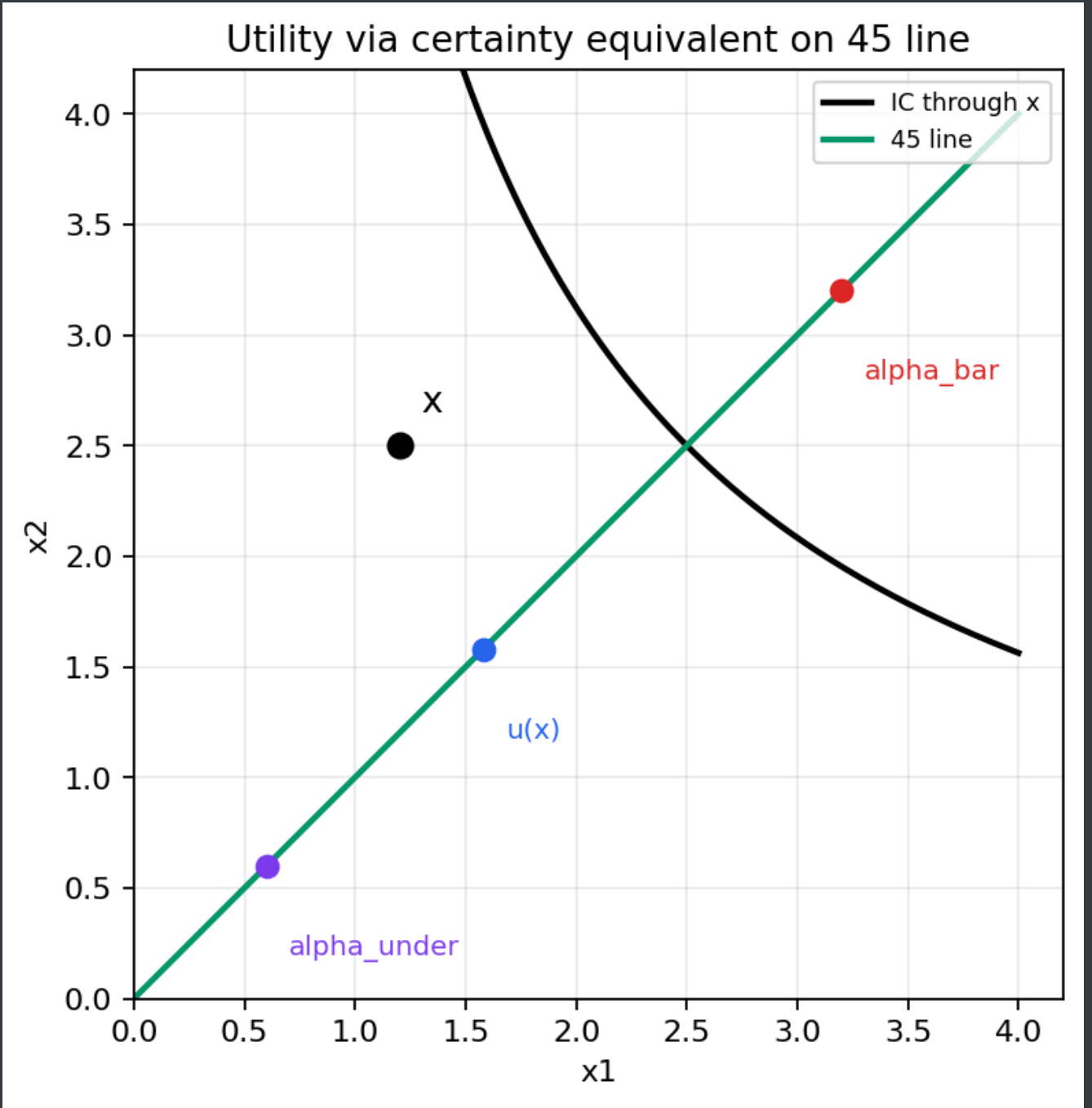
Weak monotonicity: if $x \geq y$ (componentwise), then $x \succeq y$.

Strong monotonicity: if $x \gg y$ (strictly larger in every coordinate), then $x \succ y$.

Monotonicity ensures we can compare any bundle x to points on the 45° **ray** $\alpha \mathbf{1}_n$.

Construction 4.2. Certainty-Equivalent Utility on the 45° Line

Figure (existence of $u(x)$).



Fix $x \in X$. By strong monotonicity, there exist real numbers $\underline{\alpha} < \bar{\alpha}$ such that

$$\bar{\alpha} \mathbf{1}_n \succ x \succ \underline{\alpha} \mathbf{1}_n. \quad (21)$$

Let $I_x = [\underline{\alpha}, \bar{\alpha}] \subset \mathbb{R}$ (connected). Define

$$A^+ = \{\alpha \in I_x \mid \alpha \mathbf{1}_n \succeq x\}, \quad A^- = \{\alpha \in I_x \mid x \succeq \alpha \mathbf{1}_n\}. \quad (22)$$

Step 1 - closedness. By continuity of $\succeq$, A^+ and A^- are closed (in I_x).

Step 2 - coverage. By completeness, every $\alpha \in I_x$ satisfies either $\alpha \mathbf{1}_n \succeq x$ or $x \succeq \alpha \mathbf{1}_n$, so

$$I_x \subseteq A^+ \cup A^-. \quad (23)$$

Step 3 - intersection. I_x is connected. Two closed subsets cannot cover a connected interval unless they overlap:

$$A^+ \cap A^- \neq \emptyset. \quad (24)$$

Pick $u(x) \in A^+ \cap A^-$. Then

$$x \sim u(x) \mathbf{1}_n. \quad (25)$$

Uniqueness. If $\alpha' > \alpha$ and both satisfy $x \sim \alpha \mathbf{1}_n$, strong monotonicity gives $\alpha' \mathbf{1}_n \succ \alpha \mathbf{1}_n$, contradicting transitivity unless α is unique. So $u(x)$ is well defined.

Proposition 4.3. u Represents $\succeq$

For $x, y \in X$:

$$u(x) \geq u(y) \iff x \succeq y. \quad (26)$$

Proof.

($\Rightarrow$) If $u(x) \geq u(y)$, monotonicity implies

$$u(x) \mathbf{1}_n \succeq u(y) \mathbf{1}_n. \quad (27)$$

By construction, $x \sim u(x) \mathbf{1}_n$ and $y \sim u(y) \mathbf{1}_n$. Transitivity yields $x \succeq y$.

($\Leftarrow$) If $x \succeq y$, monotonicity and transitivity imply $u(x) \mathbf{1}_n \succeq u(y) \mathbf{1}_n$, hence $u(x) \geq u(y)$ by strong monotonicity along the 45° line.

Continuity of u follows from continuity of $\succeq$ (not spelled out in the handwritten page; standard in Debreu's proof).

5. Separability of Preferences

Example 5.1. Additively Separable Form

Let $X = \mathbb{R}_{++}^n$ and $U(x) = \sum_{i=1}^n \log x_i$.

Compare bundles $(x_1, x_2, x_3, z_{-\{1,2,3\}})$ and $(y_1, y_2, y_3, z_{-\{1,2,3\}})$, where $z_{-\{1,2,3\}} \in \mathbb{R}_{++}^{n-3}$ denotes the remaining coordinates.

Let U_1 and U_2 be their corresponding utilities. If $U_1 > U_2$, then

$$\log x_1 + \log x_2 + \log x_3 > \log y_1 + \log y_2 + \log y_3. \quad (28)$$

This implies

$$(x_1, x_2, x_3, z_{-\{1,2,3\}}) \succeq (y_1, y_2, y_3, z_{-\{1,2,3\}}) \iff (x_1, x_2, x_3, w_{-\{1,2,3\}}) \succeq (y_1, y_2, y_3, w_{-\{1,2,3\}}) \quad (29)$$

for any $z_{-\{1,2,3\}}, w_{-\{1,2,3\}}$: the ranking of the first three goods is independent of the levels of the other goods.

Definition 5.2. Weak Separability

A preference relation $\succeq$ on $X \subseteq \mathbb{R}^n$ satisfies **weak separability** if for all $i \in \{1, \dots, n\}$, all $x_i, y_i \in \mathbb{R}$, and all $w_{-i}, z_{-i} \in \mathbb{R}^{n-1}$,

$$(x_i, z_{-i}) \succeq (y_i, z_{-i}) \iff (x_i, w_{-i}) \succeq (y_i, w_{-i}). \quad (30)$$

Interpretation. The ranking between two bundles that differ only in coordinate i does not depend on the fixed levels of the other coordinates.

Definition 5.3. Strong Separability

A preference relation $\succeq$ on $X \subseteq \mathbb{R}^n$ satisfies **strong separability** if for all subsets $I \subset \{1, \dots, n\}$ and all $w_{-I}, z_{-I} \in \mathbb{R}^{n-|I|}$,

$$\left((x_i)_{i \in I}, z_{-I} \right) \succeq \left((y_i)_{i \in I}, z_{-I} \right) \iff \left((x_i)_{i \in I}, w_{-I} \right) \succeq \left((y_i)_{i \in I}, w_{-I} \right). \quad (31)$$

Interpretation. The ranking over any subset I of goods is independent of the levels of goods outside I . Strong separability implies weak separability.

Proposition 5.4. Representation of Weakly Separable Preferences

Suppose $\succeq$ on $\mathbb{R}^n$ is rational, continuous, and weakly separable. Then there exist functions $u_i : \mathbb{R} \rightarrow \mathbb{R}$ for each i and an aggregator f such that

$$U(x) = f(u_1(x_1), \dots, u_n(x_n)) \quad (32)$$

represents $\succeq$.

This is the useful direction for the notes: weak separability lets us construct coordinate-level sub-utilities and then aggregate them. The converse requires additional restrictions on the aggregator f ; an arbitrary nested expression $f(u_1, \dots, u_n)$ need not make one-coordinate comparisons independent of the other coordinates.

Proof idea.

1. For each i , define $\succeq_i$ on $\mathbb{R}$ by: $x_i \succeq_i y_i$ if and only if $(x_i, z_{-i}) \succeq (y_i, z_{-i})$ for some (hence any, by weak separability) z_{-i} .
2. Weak separability ensures each $\succeq_i$ is well defined. By rationality and continuity of $\succeq$, each $(\succeq_i)_{i=1}^n$ is rational and continuous.
3. Hence there exist continuous utility functions $u_i : \mathbb{R} \rightarrow \mathbb{R}$ representing each $\succeq_i$, and a continuous $U : \mathbb{R}^n \rightarrow \mathbb{R}$ representing $\succeq$.
4. Define $f(v_1, \dots, v_n) \equiv U(x_1, \dots, x_n)$ whenever $v_i = u_i(x_i)$ for each i . For any $v \in \prod_{i=1}^n u_i(\mathbb{R})$, pick x with $v_i = u_i(x_i)$; then $f(v_1, \dots, v_n)$ is well defined and independent of the choice of x (by weak

separability).

Remark. With **monotonicity** added, one can take each u_i and f to be monotonic as well.

Intuitive Proof of Proposition 5.4: A Three-Step Logic

The core idea behind why weak separability guarantees a nested utility structure $U(x) = f(u_1(x_1), \dots, u_n(x_n))$ can be broken down into three logical steps.

Step 1 – Isolating Local Preferences (Defining $\succeq_i$)

By the definition of weak separability, when you compare two different consumption levels of the i -th good (say, x_i and y_i), the quantities of all other goods do not affect your ranking.

- Because the external environment is irrelevant to this specific choice, we can completely strip away the other goods.
- This allows us to establish an isolated, self-consistent local preference relation on the choice space of good i alone, denoted as $\succeq_i$.

Step 2 – Constructing Local Scales (Introducing Sub-utility u_i)

Since the global preference relation $\succeq$ is rational and continuous, each isolated local preference relation $\succeq_i$ inherits these properties (rationality and continuity).

- By **Debreu's representation theorem** (Proposition 3.6), any rational and continuous preference relation can be represented by a continuous real-valued function.
- Consequently, every single good i obtains its own continuous sub-utility function, denoted as $u_i(x_i)$.

Step 3 – Eliminating Ambiguity (Validating the Aggregator f)

The final and critical step is to prove that if two entirely different bundles yield identical sub-utility scores across every single dimension, they must also yield the exact same total utility.

Suppose there are two bundles x and y such that $u_1(x_1) = u_1(y_1)$, $u_2(x_2) = u_2(y_2)$, ..., and $u_n(x_n) = u_n(y_n)$. By definition, this implies local indifference in each dimension: $x_1 \sim_1 y_1$, $x_2 \sim_2 y_2$, ..., $x_n \sim_n y_n$.

Utilizing weak separability, we can substitute the components of bundle x with those of bundle y one item at a time without changing the global preference ranking:

$$(x_1, x_2, \dots, x_n) \sim (y_1, x_2, \dots, x_n) \sim (y_1, y_2, \dots, x_n) \sim \dots \sim (y_1, y_2, \dots, y_n). \quad (33)$$

At the end of this chain of substitutions, the global bundle x is completely indifferent to bundle y , so their total utility must be equal: $U(x) = U(y)$.

Conclusion. The global utility U depends *solely and uniquely* on the vector of sub-utility scores $(u_1, \dots, u_n)$. Therefore, there must exist a well-defined aggregator function f such that

$$U(x) = f(u_1(x_1), \dots, u_n(x_n)). \quad (34)$$

Proposition 5.5. Additively Separable Representation

Suppose $\succeq$ on $\mathbb{R}^n$ is rational, continuous, strongly separable, and satisfies the standard regularity assumptions used in Debreu's additive-separability theorem. Then there exist functions $u_i : \mathbb{R} \rightarrow \mathbb{R}$ for $i = 1, \dots, n$ such that

$$U(x) = \sum_{i=1}^n u_i(x_i) \quad (35)$$

represents $\succeq$.

Proof idea (for $n = 3$).

Suppose $n = 3$. By strong separability, define sub-relations $(\succeq_I)$ on $\mathbb{R}^{|I|}$ for each $I \subset \{1, 2, 3\}$. By rationality and continuity, each can be represented by a utility function:

- $U_{12} : \mathbb{R}^2 \rightarrow \mathbb{R}$ represents $\succeq_{\{1,2\}}$,
- $U_{23} : \mathbb{R}^2 \rightarrow \mathbb{R}$ represents $\succeq_{\{2,3\}}$,
- $U : \mathbb{R}^3 \rightarrow \mathbb{R}$ represents $\succeq$.

By weak separability (which follows from strong separability), on $U_{12}(\mathbb{R}^2) \times u_3(\mathbb{R})$:

$$\tilde{U}(x_1, x_2, x_3) = F(U_{12}(x_1, x_2), u_3(x_3)), \quad (36)$$

where $F : U_{12}(\mathbb{R}^2) \times u_3(\mathbb{R}) \rightarrow \mathbb{R}$. Similarly, on $u_1(\mathbb{R}) \times U_{23}(\mathbb{R}^2)$:

$$\tilde{U}(x_1, x_2, x_3) = H(u_1(x_1), U_{23}(x_2, x_3)), \quad (37)$$

where $H : u_1(\mathbb{R}) \times U_{23}(\mathbb{R}^2) \rightarrow \mathbb{R}$.

Decompose the sub-utilities:

$$U_{12}(x_1, x_2) = G(u_1(x_1), u_2(x_2)), \quad U_{23}(x_2, x_3) = K(u_2(x_2), u_3(x_3)). \quad (38)$$

Combining these four equations yields the functional equation

$$F(G(u_1, u_2), u_3) = H(u_1, K(u_2, u_3)). \quad (39)$$

A standard solution (Aczel-type functional equation) is

$$F(v_{12}, u_3) = \phi(\phi_{12}^{-1}(v_{12}) + \phi_3(u_3)), \quad (40)$$

$$G(u_1, u_2) = \phi_{12}(\phi_1(u_1) + \phi_2(u_2)), \quad (41)$$

$$H(u_1, v_{23}) = \phi(\phi_1(u_1) + \phi_{23}^{-1}(v_{23})), \quad (42)$$

$$K(u_2, u_3) = \phi_{23}(\phi_2(u_2) + \phi_3(u_3)). \quad (43)$$

Plugging K into H gives

$$\tilde{U}(x_1, x_2, x_3) = \phi(\phi_1(u_1) + \phi_2(u_2) + \phi_3(u_3)). \quad (44)$$

Setting $U(x_1, x_2, x_3) \equiv \phi^{-1}(\tilde{U})$ yields the additive form $U(x) = \sum_{i=1}^3 u_i(x_i)$ (after redefining the u_i via the ϕ_i). The argument extends to general n .

6. Choice Theory and Revealed Preference

Definition 6.1. Choice Problem

A **choice problem** is a pair $(\mathcal{B}, C)$, where:

- X : finite set of alternatives,
- $\mathcal{B}$: **domain of choice**, with $\mathcal{B} \subseteq 2^X \setminus \{\emptyset\}$,
- C : **choice correspondence**, $C : \mathcal{B} \rightarrow 2^X$, such that $C(A) \subseteq A$ for every $A \in \mathcal{B}$.

Note. The choice correspondence C is **observable**, but the underlying preference relation $\succeq$ is **unobservable**. Can we recover $\succeq$ from $(\mathcal{B}, C)$?

Definition 6.2. Rationalization

We say that a preference relation $\succeq$ on X **rationalizes** the choice correspondence C if for every menu $A \in \mathcal{B}$,

$$C(A) = C^*(A, \succeq) = \{x \in A \mid x \succeq y, \forall y \in A\}. \quad (45)$$

That is, chosen elements are exactly the $\succeq$ -maximal elements of each menu.

Definition 6.3. Weak Axiom of Revealed Preference (WARP)

A choice structure $(\mathcal{B}, C)$ satisfies the **Weak Axiom of Revealed Preference** (WARP) if for all $x, y \in X$ and all menus $A, B \in \mathcal{B}$:

If $x, y \in A$, $x \in C(A)$, $x, y \in B$, and $y \in C(B)$, then $x \in C(B)$.

Interpretation. If x is chosen when y is available, and y is chosen from another menu that also contains x , then x must also be chosen from that menu - otherwise we would observe inconsistent revealed preferences.

Proposition 6.4. Rationalizability and WARP

Suppose $\mathcal{B}$ is **rich** (i.e., $\mathcal{B}$ contains every nonempty subset of X). Then C is rationalizable if and only if $(\mathcal{B}, C)$ satisfies WARP.

Proposition 6.5. Equivalent Conditions for Rationalizability

Suppose $\mathcal{B} = 2^X \setminus \{\emptyset\}$. The following conditions are equivalent (each is a behaviorally testable equivalent to WARP and full rationalizability):

1. **WARP:** as in Definition 6.3.
2. **Houthakker's Axiom:** If there are menus $A, B \in \mathcal{B}$ such that $x, y \in A \cap B$, $x \in C(A)$, and $y \in C(B)$, then $x \in C(B)$ and $y \in C(A)$.

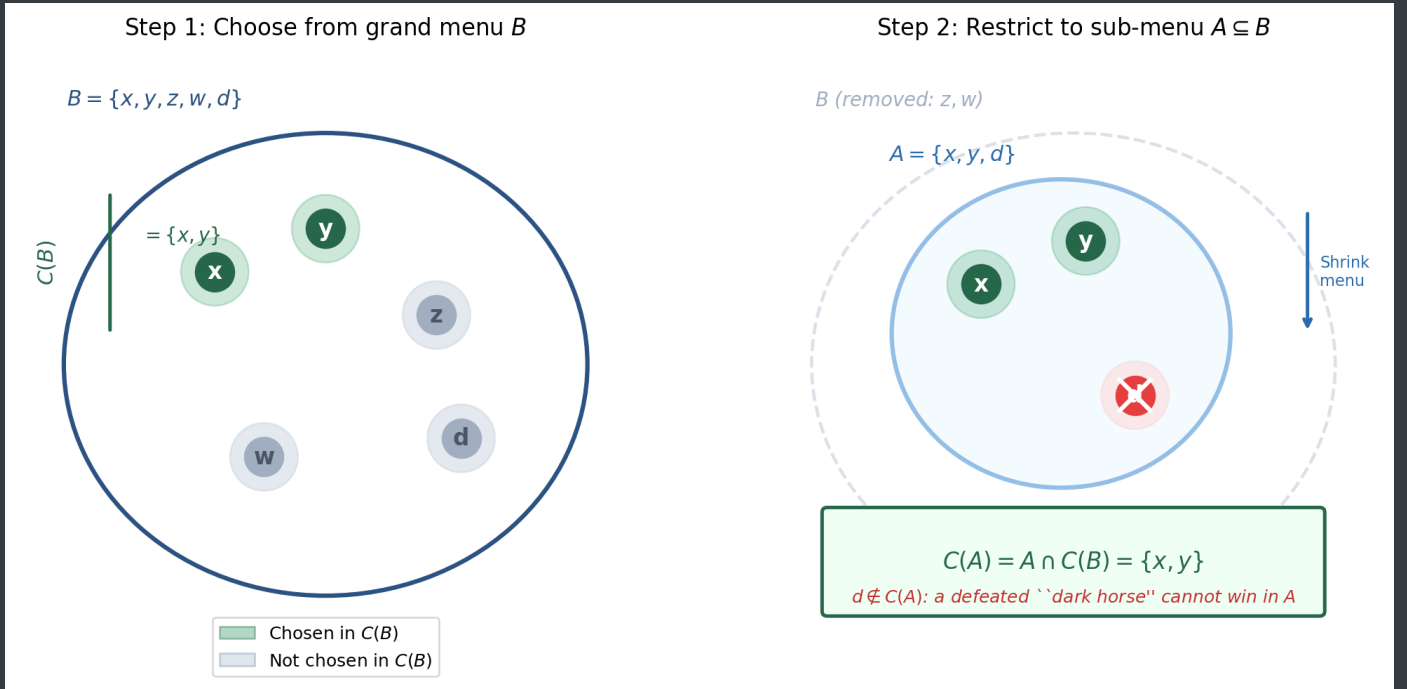
Intuition (translational invariance of choice). Choice behavior must remain perfectly consistent when moving across overlapping menus. If two alternatives x and y are simultaneously available in both menu A and menu B , their relative standing is locked: if x is chosen in A and y is chosen in B , then both must be chosen in both menus. They share the same fate - the mere presence of other, irrelevant alternatives in A or B cannot break their tie.

3. **Arrow's Axiom:** For any menus $A, B \in \mathcal{B}$ with $A \subseteq B$, if $A \cap C(B) \neq \emptyset$, then

$$C(A) = A \cap C(B). \quad (46)$$

Intuition (independence of irrelevant alternatives). This axiom governs how choice sets contract when a large menu shrinks to a smaller sub-menu. Suppose you select a set of optimal elements $C(B)$ from a grand menu B . If the menu is restricted to a smaller subset $A \subseteq B$, and some of the original winners from B are still present in A (i.e., $A \cap C(B) \neq \emptyset$), then the new choice set $C(A)$ must consist **strictly and exclusively** of those surviving grand winners. Shrinking the menu can never cause a previously defeated alternative - a "dark horse" eliminated in B - to suddenly emerge as a winner in A . Choices on the sub-menu are independent of irrelevant alternatives that were removed.

Figure (Arrow's Axiom: menu contraction).



Chapter 3. Voting: Preference Aggregation

7. Preference Aggregation

Definition 7.1. Social Welfare Function

Let $N = \{1, \dots, n\}$ be the set of individuals and X a finite set of alternatives.

- $\succeq_i$: a rational preference relation on X for individual i .
- $\mathcal{R}$: the set of all rational preference relations on X .
- $\mathcal{R}_i \subseteq \mathcal{R}$: the set of admissible preferences for individual i .
- $\prod_{i \in N} \mathcal{R}_i$: the domain of **preference profiles**.

Let $P \subseteq \mathcal{R}$ denote strict preference. A **social welfare function (SWF)** is a mapping

$$F : \prod_{i \in N} \mathcal{R}_i \longrightarrow \mathcal{R}, \quad (47)$$

which assigns a collective preference $\succeq = F(\succeq_1, \dots, \succeq_n)$ to each profile $(\succeq_1, \dots, \succeq_n)$.

Example 7.2. Binary Choice and Majority Rule

Suppose $|X| = 2$ and n is odd with $n \geq 3$. Identify the two alternatives with $\{x, y\}$ and encode individual votes by

$$X = \{1, -1\}, \quad x_i = 1 \iff x \succ_i y, \quad x_i = -1 \iff y \succ_i x. \quad (48)$$

The aggregation problem becomes

$$\bar{F} : \{1, -1\}^n \longrightarrow \{1, -1\}. \quad (49)$$

The **majority rule** $\bar{F}^M$ is defined by, for $x = (x_1, \dots, x_n)$,

$$\bar{F}^M(x) = \begin{cases} 1, & \text{if } \sum_{i \in N} x_i > 0, \\ -1, & \text{if } \sum_{i \in N} x_i < 0. \end{cases} \quad (50)$$

Since n is odd, $\sum_{i \in N} x_i \neq 0$, so ties cannot occur.

8. May's Theorem

Work in the binary setting with $X = \{1, -1\}$, an odd number of voters, and a rule $F : \{1, -1\}^n \rightarrow \{1, -1\}$.

Definition 8.1. Axioms for Binary Aggregation

1. **Pareto:** if $x = (1, \dots, 1)$, then $F(x) = 1$; if $x = (-1, \dots, -1)$, then $F(x) = -1$.
2. **Anonymity:** for every permutation $\pi : N \rightarrow N$ and every $x \in \{1, -1\}^n$,

$$F(x) = F(\pi \circ x), \quad (\pi \circ x)_i = x_{\pi(i)}. \quad (51)$$

3. **Neutrality:** for every $x \in \{1, -1\}^n$,

$$F(-x) = -F(x). \quad (52)$$

4. **Positive responsiveness:** for all $x, x' \in \{1, -1\}^n$ with $x \geq x'$ coordinatewise, if $F(x') = 1$, then $F(x) = 1$.

Example 8.2. Majority Rule for $n = 3$

The table below lists all profiles and the output of the majority rule $\bar{F}^M$.

x_1	x_2	x_3	$\bar{F}^M(x)$
1	1	1	1
1	1	-1	1
1	-1	1	1
-1	1	1	1
-1	-1	1	-1
-1	1	-1	-1
1	-1	-1	-1
-1	-1	-1	-1

Theorem 8.3. May's Theorem

A rule $F : \{1, -1\}^n \rightarrow \{1, -1\}$ with odd n satisfies **Pareto**, **Anonymity**, **Neutrality**, and **Positive responsiveness** if and only if F is the **majority rule**.

Proof idea (for $n = 3$).

x_1	x_2	x_3	$F(x)$
1	1	1	1
1	1	-1	?
1	-1	1	?
-1	1	1	?
-1	-1	1	?
-1	1	-1	?
1	-1	-1	?
-1	-1	-1	-1

- **Pareto** pins down the all-1 and all- -1 rows.
- **Anonymity** forces the four profiles with two 1's and one -1 to share the same outcome, and likewise for the profiles with one 1 and two -1 's.
- **Neutrality** links each row to its negation: if $F(1, 1, -1) = 1$, then $F(-1, -1, 1) = -1$.
- **Positive responsiveness** then propagates outcomes upward: once a two-1 profile is chosen as 1, every profile with more 1's must also be chosen as 1; any deviation from majority rule creates a **responsiveness contradiction**.

Hence every undetermined entry is forced to match $\bar{F}^M$.

9. Preference Cycles and Scoring Rules

When $|X| \geq 3$, pairwise majority voting need not yield a transitive social ranking: a **cycle** may arise.

Example 9.1. Condorcet Cycle

Let $X = \{x, y, z\}$ and consider three voters with preferences

Voter	Preference
1	$x \succ_1 y \succ_1 z$
2	$y \succ_2 z \succ_2 x$
3	$z \succ_3 x \succ_3 y$

Pairwise majority comparisons give

$$x \succ y, \quad y \succ z, \quad z \succ x, \quad (53)$$

so no alternative is a Condorcet winner and the social relation is cyclic.

Definition 9.2. Scoring Rule

Fix a score function $S(\cdot, \succeq_i)$ such that

$$S(x, \succeq_i) > S(y, \succeq_i) \iff x \succ_i y. \quad (54)$$

For a profile $\succeq = (\succeq_1, \dots, \succeq_n)$, define the **total score**

$$S(x, \succeq) = \sum_{i \in N} S(x, \succeq_i), \quad (55)$$

and the induced social preference

$$x R^S(\succeq) y \iff S(x, \succeq) \geq S(y, \succeq). \quad (56)$$

Example 9.3. Borda Count

With $|X| = 3$, assign Borda points 3 (1st), 2 (2nd), and 1 (3rd). Consider 5 voters split into two groups:

Rank	3 voters	2 voters	Borda points
1st	y	x	3
2nd	x	z	2
3rd	z	y	1

Total scores:

$$S(x, \succeq) = 3 \cdot 2 + 2 \cdot 3 = 12, \quad S(y, \succeq) = 3 \cdot 3 + 2 \cdot 1 = 11, \quad S(z, \succeq) = 3 \cdot 1 + 2 \cdot 2 = 7. \quad (57)$$

Hence the Borda social ranking is

$$x R^B(\succeq) y R^B(\succeq) z. \quad (58)$$

10. Voting Rules: Examples and Comparisons

Example 10.1. Borda Scores Can Change Without Changing x vs y

Consider a different profile with the same split between 3 and 2 voters:

Rank	3 voters	2 voters
1st	y	x
2nd	z	z
3rd	x	y

Borda totals (3-2-1 points):

$$S(x, \succeq) = 3 \cdot 1 + 2 \cdot 3 = 9, \qquad S(y, \succeq) = 3 \cdot 3 + 2 \cdot 1 = 11, \qquad S(z, \succeq) = 3 \cdot 2 + 2 \cdot 2 = 10, \quad (59)$$

so

$$yR^B(\succeq)zR^B(\succeq)x. \tag{60}$$

Remark. In both Example 9.3 and Example 10.1, 3 voters rank $y \succ x$ while 2 voters rank $x \succ y$. Yet merely changing the ranks of z and x (holding the x - y comparison fixed within each group) reverses the Borda winner from x to y . This illustrates sensitivity to "irrelevant" alternatives.

Example 10.2. Borda, Plurality, and the Condorcet Winner

Now consider 19 voters divided into five groups:

Rank	4 voters	3 voters	6 voters	2 voters	4 voters
1st	x	x	y	z	y
2nd	y	z	z	x	z
3rd	z	y	x	y	x

Plurality rule (F^P): only first-place votes count (1 point each).

Alternative	First-place votes
x	$4 + 3 = 7$
y	$6 + 4 = 10$
z	2

So F^P selects y first and x second.

Borda rule (F^B): assign 3-2-1 points by rank.

Alternative	Borda calculation	Total
x	$4 \cdot 3 + 3 \cdot 3 + 6 \cdot 1 + 2 \cdot 2 + 4 \cdot 1$	35
y	$4 \cdot 2 + 3 \cdot 1 + 6 \cdot 3 + 2 \cdot 1 + 4 \cdot 3$	43
z	$4 \cdot 1 + 3 \cdot 2 + 6 \cdot 2 + 2 \cdot 3 + 4 \cdot 2$	36

So F^B ranks y first, z second, and x third.

Condorcet winner. Pairwise majority comparisons give

- x vs y : y beats x (10 voters prefer y to x ; 9 prefer x to y).
- y vs z : y beats z (14 voters prefer y to z ; 5 prefer z to y).

Since y defeats every other alternative head-to-head, y is the **Condorcet winner**.

Takeaway. Different aggregation rules can disagree even on the same profile; here plurality, Borda, and the Condorcet criterion all select y , but the full social rankings need not coincide.

11. Independence of Irrelevant Alternatives and Arrow's Theorem

Definition 11.1. Independence of Irrelevant Alternatives (IIA)

For a SWF F , say that F satisfies **IIA** if for all profiles $\succeq, \succeq' \in \prod_{i \in N} \mathcal{R}_i$ and all $x, y \in X$,

$$\succeq_i \upharpoonright_{\{x,y\}} = \succeq'_i \upharpoonright_{\{x,y\}} \quad \text{for all } i \in N \implies (xF(\succeq)y \iff xF(\succeq')y). \quad (61)$$

That is, the social ranking of x and y depends only on individual rankings of x and y , not on how individuals rank other alternatives.

Definition 11.2. Pareto Condition

F satisfies **(Weak) Pareto** if whenever $x \succ_i y$ for **every** $i \in N$, then $xF(\succeq)y$.

Correction from handwritten notes. The slide wrote "for any $i \in N$ "; the standard Pareto axiom requires unanimity ("for all $i \in N$ ").

Definition 11.3. Dictatorship

F is **dictatorial** if there exists $i^* \in N$ such that for every profile $\succeq \in \prod_{i \in N} \mathcal{R}_i$ and all $x, y \in X$,

$$x \succ_{i^*} y \implies xF(\succeq)y. \quad (62)$$

Individual i^* is a **dictator**: his strict preference always determines the social ranking.

Theorem 11.4. Arrow's Theorem (Lecture Formulation)

Assume $|X| \geq 3$ and the domain is the full set of rational preference profiles. A SWF F satisfies **IIA** and **Pareto** if and only if F is **dictatorial**.

Interpretation. Once we require social rankings to respect unanimous strict preference and to depend only on pairwise comparisons, the only SWFs left are dictatorships. This is the lecture's streamlined version of Arrow's impossibility logic; the full theorem adds **Universal Domain** and **Non-Dictatorship** and concludes that no SWF can satisfy all four conditions simultaneously when $|X| \geq 3$.

12. Arrow's Theorem: Proof via Decisive Groups

The proof of Arrow's Theorem presented here is a classic construction known as the **Proof via Decisive Groups**. The core intuition can be understood as "**cell division and power contraction**": we first prove that the entire society as a collective group is omnipotent (decisive); then we show that if any group holds this absolute power, a strictly smaller subgroup within it must also inherit that exact same power; by repeatedly shrinking the power circle, we eventually lock it down to a single individual - the **dictator**.

Core Concepts: Understanding "Decisive Groups"

Before diving into the steps, we must clearly distinguish between the levels of political "power" defined in the text:

Let $F : \mathcal{R}^n \rightarrow \mathcal{R}$ be a SWF, and let $\emptyset \neq G \subseteq N$.

- **$\{x, y\}$ -decisive Group:**

If every member inside G strictly prefers $x \succ y$, and every outsider (G^c) strictly prefers $y \succ x$, and the final social preference still yields $x F(\succ) y$, then G is $\{x, y\}$ -decisive.

In Plain English: Even if the entire rest of the world stands against us, as long as our group G unites to support x over y , society must rule in our favor.

- **$\{x, y\}$ -completely decisive Group:**

An upgrade in power. As long as everyone inside G prefers $x \succ y$, **no matter what outsiders think or how they rank the alternatives**, the social preference is guaranteed to be strictly $x P(\succ) y$.

- **Completely Decisive Group:**

The ultimate authority. If a group G is completely decisive for **every single pair** of alternatives (whether it is x vs y , a vs b , or z vs x), it is a *completely decisive group*. If this group shrinks to a single individual $\{i^*\}$, that person is a **dictator**.

Step 1 – Field Expansion Lemma

Core Conclusion (Proposition 12.2): "*Local power implies global power.*" If a group G is completely decisive over just *one* specific pair of alternatives $\{x, y\}$, it automatically becomes completely decisive over *all* possible pairs $\{a, b\}$ in the entire choice space.

Detailed Logic (Taking Case 1 as the prime example). We want to prove: Given that G completely controls the outcome of x vs y , how do we show G controls an entirely unrelated pair a vs b ?

The proof constructs a highly specific and clever preference profile:

- Let all individuals **inside** G rank: $a \succ_i x \succ_i y \succ_i b$
- Let all individuals **outside** G (G^c) rank: $a \succ_j x$ and $y \succ_j b$ (their relative rankings of a vs b and x vs y can be anything, as long as these two boundaries hold)

Now, like a row of falling dominoes, we use the **Pareto Principle** and **Transitivity** to derive the social preference step-by-step:

1. **Deducing a vs x :** Notice that *everyone* in society (both inside G and outside G) agrees that $a \succ x$. By the **Pareto Principle**, society must rank:

$$a P(\succ) x$$

2. **Deducing x vs y :** Everyone inside G prefers $x \succ y$. Since our initial assumption is that G is $\{x, y\}$ -completely decisive, society *must* side with G regardless of the outsiders:

$$x P(\succ) y$$

3. **Deducing y vs b :** Notice again that *everyone* in society agrees that $y \succ b$. By the **Pareto Principle**, society must rank:

$$y P(\succ) b$$

4. **The Chain of Transitivity:** Stringing these three social preferences together (a beats x , x beats y , and y beats b), the **Transitivity** of rational social preferences forces:

$$a P(\succ) b$$

The Role of IIA (Independence of Irrelevant Alternatives):

You might object: "*This social outcome only holds under this specific, artificially engineered profile!*" This is where **IIA** comes in. IIA dictates that society's ranking of a vs b depends **strictly and exclusively** on how individuals rank a vs b - the positions of x and y are completely irrelevant.

In our construction, members inside G strictly preferred $a \succ b$, while the preferences of outsiders for a vs b were left completely unrestricted. Yet, society was forced to conclude $a P(\succ) b$. This proves that whenever G universally prefers $a \succ b$, society will always rule $a P(\succ) b$. Thus, G is $\{a, b\}$ -completely decisive.

Cases 2-3 (sketch). The remaining cases handle overlapping pairs:

- **Case 2:** $\{a, b\}$ shares one element with $\{x, y\}$. Two symmetric subcases - e.g., to prove $\{x, z\}$ -decisiveness, construct $G : x \succ y \succ z$, $G^c : y \succ z \succ x$, then chain xPy (decisiveness), yPz (Pareto), xPz (transitivity). Upgrade to completely decisive via IIA.

- **Case 3:** $\{a, b\} = \{x, y\}$, trivial by hypothesis.

Thus G is $\{a, b\}$ -completely decisive for all $a, b \in X$. $\square$

Step 2 – Group Splitting Lemma

Core Conclusion (Proposition 12.3): "Power must inevitably concentrate into a smaller core faction." If a group G containing at least 2 people is completely decisive, we can split G into two smaller subgroups, and **exactly one** of those subgroups will inherit the full, completely decisive power.

Detailed Logic. We split the completely decisive group G arbitrarily into two non-empty, disjoint subsets: G_1 and G_2 . We introduce a third alternative z (requiring at least 3 alternatives in total) and construct the following preference profile:

Group	Individual Preference Ranking
Subgroup G_1	$x \succ y \succ z$
Subgroup G_2	$z \succ x \succ y$
Outsiders (G^c)	$y \succ z \succ x$

Let's dissect the social choices forced by this setup:

1. First, look at x **vs** y :

Every single member of G_1 prefers $x \succ y$, and every single member of G_2 prefers $x \succ y$. Therefore, **everyone in the larger group G** prefers $x \succ y$. Since G is known to be completely decisive, society must rule:

$$x P(\succ) y$$

2. Next, we look at how society handles z **vs** y . By completeness and rationality, there are only two possibilities for the social preference:

- **Subcase (i): Society rules $z P(\succ) y$**

Look closely at the table: **who** in this world actually wants $z \succ y$? *Only G_2 .* G_1 opposes it ($y \succ z$), and the outsiders oppose it ($y \succ z$).

Despite facing universal opposition, G_2 's preference prevailed. By **IIA**, this means G_2 is $\{z, y\}$ -decisive. Invoking Step 1 (Field Expansion), G_2 immediately upgrades to being a **completely decisive group**.

- **Subcase (ii): Society rules $y R(\succ) z$ (weakly prefers y to z)**

If subcase (i) is false, then subcase (ii) must be true. Now we have two pieces of social data: $x P(\succ) y$ (from step 1) and $y R(\succ) z$.

By **Transitivity**, a strict winner (x) beating a weak winner (y) means x must strictly beat z :

$$x P(\succ) z$$

Look back at the table: **who** in this world actually wants $x \succ z$? *Only G_1 .* G_2 opposes it ($z \succ x$), and the outsiders oppose it ($z \succ x$).

Once again, a lone subgroup wins against the world. This means G_1 is $\{x, z\}$ -decisive. Invoking Step 1, G_1 upgrades to being a **completely decisive group**.

Summary: No matter how society rules on the margin, the baton of absolute power is cleanly passed down to either G_1 or G_2 . $\square$

Step 3 – The Birth of a Dictator

With the Field Expansion and Group Splitting lemmas in our arsenal, the final phase of Arrow's Theorem becomes a straightforward game of "peeling the onion":

1. **Planting the Seed of Power:** By the **Pareto Principle**, if every single person in the entire nation (N) agrees that $x \succ y$, society must choose $x P(\succ) y$. This perfectly satisfies the definition of a completely decisive group. Thus, our starting grand coalition, $G_0 = N$, is **completely decisive**.
2. **Continuous Slicing:**
 - If society consists of only 1 person, he is trivially the dictator.
 - If $|N| \geq 2$, we apply **Step 2 (Group Splitting Lemma)**. We split N and extract a strictly smaller, proper subgroup G_1 that retains complete decisiveness.
 - If G_1 still contains more than one person, we split it again to find an even smaller completely decisive subgroup G_2 .
 - This creates a strictly shrinking chain of absolute power:
$$N = G_0 \supset G_1 \supset G_2 \supset \dots$$
3. **The Termination Point:** Because the total number of individuals in society (N) is finite, this shrinking chain cannot go on forever. It is mathematically forced to terminate at a singleton set containing exactly one person:
$$G_k = \{i^*\}$$
4. **Crowning the Dictator:** By definition, because the single-person group $\{i^*\}$ is completely decisive, whatever i^* strictly prefers will become the absolute social choice, completely ignoring the preferences of the remaining $n - 1$ citizens.

Individual i^* is the **Dictator**, completing the logical proof of Arrow's Theorem. $\square$

13. Single-Peaked Preferences and the Median Voter Theorem

When the domain of preferences is restricted to **single-peaked** preferences, positive aggregation results become possible - Arrow's impossibility no longer applies.

Definition 13.1. Single-Peaked Preferences

Let $X \subseteq \mathbb{R}$ be a linearly ordered set of alternatives. A strict preference $\succ_i$ on X is **single-peaked** if there exists a **peak** $t_i \in X$ such that:

- For all $x, y \in X$ with $x < y \leq t_i$: $y \succ_i x$ (increasing toward the peak).
- For all $x, y \in X$ with $t_i \leq x < y$: $x \succ_i y$ (decreasing away from the peak).

Interpretation. Each voter has a most-preferred point; the further an alternative lies from that peak in either direction, the less the voter likes it.

Let $\mathcal{SP}$ denote the set of single-peaked preferences, and $\mathcal{SP}^n$ the domain of single-peaked profiles with n voters.

Example 13.2. Single-Peaked Peaks on a Line

Suppose $n = 7$ (odd) and the peaks are

$$t_{(1)} < t_{(2)} < t_{(3)} < t_{(4)} < t_{(5)} < t_{(6)} < t_{(7)}, \quad (63)$$

where $t_{(k)}$ denotes the k -th smallest peak. The median peak is $t_{(4)}$.

Proposition 13.3. Median Voter = Condorcet Winner

Assume n is odd, $n = 2m + 1$. Let $t_{(1)} \leq t_{(2)} \leq \dots \leq t_{(n)}$ be the ordered peaks. Then the **median peak** $t_{(m+1)}$ is a **Condorcet winner**: for any alternative $y \neq t_{(m+1)}$, a majority strictly prefers $t_{(m+1)}$ to y .

Proof sketch. Take any $y < t_{(m+1)}$. At least $m + 1$ voters have peaks $t_{(m+1)}$ or higher. By single-peakedness, all such voters prefer $t_{(m+1)}$ to y (since y lies to the left of their peaks). Hence a majority prefers $t_{(m+1)}$ to y . Symmetric argument for $y > t_{(m+1)}$. $\square$

Definition 13.4. Median Voter Rule

The **median voter rule** is the social choice function $f : \mathcal{SP}^n \rightarrow X$ defined by

$$f(\succ_1, \dots, \succ_n) = \text{median}\{t_1, \dots, t_n\}, \quad (64)$$

where t_i is the peak of voter i 's single-peaked preference.

Theorem 13.5. Characterization of the Median Voter Rule

On the single-peaked domain $\mathcal{SP}^n$, strategy-proof social choice rules are characterized by **generalized median rules**. If n is odd and the rule is also anonymous and Pareto efficient, the rule reduces to the ordinary **median voter rule**.

Remark. The generalized median rule extends the median voter concept by allowing fixed "phantom voters" (additional fixed peaks) that bias the outcome. The pure median voter rule is the special case in which the outcome is the median of the actual voters' peaks.

Chapter 4. Social Welfare Ordering

14. Social Welfare Functionals and the Welfarism Theorem

Arrow's framework (Sections 11-12) works with **ordinal** preference profiles. We now allow **cardinal** (utility) information by defining aggregation rules on utility profiles rather than preference relations.

Definition 14.1. Social Welfare Functional (SWFL)

Let X be a set of social alternatives and $N = \{1, \dots, n\}$ the set of individuals.

- A **utility profile** is a function $U : X \rightarrow \mathbb{R}^n$, where $U(x) = (u_1(x), \dots, u_n(x))$ assigns each individual's utility value to alternative x . The set of all possible utility profiles is denoted $\mathcal{U}$.
- Let $D \subseteq \mathcal{U}$ be the **domain** of admissible utility profiles.

A **Social Welfare Functional (SWFL)** is a mapping

$$F : D \longrightarrow \mathcal{R}, \tag{65}$$

where $\mathcal{R}$ is the set of all rational (complete and transitive) preference relations on X . For any profile $U \in D$, we write

$$R_U = F(U), \qquad xR_Uy \iff x \text{ is socially at least as good as } y \text{ under } U. \tag{66}$$

The associated strict social preference and social indifference are denoted P_U and I_U , respectively.

Intuition. An SWFL produces a social ranking for each profile of individual utility functions. This is more general than Arrow's SWF because it admits cardinal information.

Example 14.2. Two Individuals, Three Alternatives

Let $N = \{1, 2\}$ and $X = \{x, y, z\}$. Consider two utility profiles U and V :

	x	y	z
U	$(2, 1)$	$(3, 5)$	$(1, 2)$
V	$(2, 1)$	$(-1, -1)$	$(1, 2)$

Under profile U , both individuals agree y is best but disagree on x vs z (individual 1 prefers x , individual 2 prefers z). Under profile V , the individually worst alternative y drags down relative to x and z .

14.1 Axioms for Social Welfare Functionals

Definition 14.3. Universal Domain (UD)

The domain D is unrestricted: $D = \mathcal{U}$, the set of *all* possible utility profiles.

Definition 14.4. Binary Independence (BI)

For any $x, y \in X$ and any $U, V \in D$,

$$U(x) = V(x) \text{ and } U(y) = V(y) \implies (xR_U y \iff xR_V y). \quad (67)$$

Interpretation. The social ranking of x and y depends only on how much utility each individual derives from x and y - not on utility values attached to other alternatives. BI is the utility-space analogue of Arrow's IIA.

Definition 14.5. Pareto Indifference (PI)

For any $x, y \in X$ and any $U \in D$,

$$U(x) = U(y) \implies xI_U y. \quad (68)$$

If every individual is indifferent between x and y (same utility vector), society must also be indifferent. This is a weak form of the Pareto principle applied to the utility framework.

Example. In the table of Example 14.2, suppose a third profile U' assigns $(2, 1)$ to both x and w . Then by PI: $xI_{U'} w$.

Definition 14.6. Strong Neutrality (SN)

For any $x, y, z, w \in X$ and any $U, V \in D$,

$$U(x) = V(z) \text{ and } U(y) = V(w) \implies (xR_U y \iff zR_V w). \quad (69)$$

Interpretation. SN says: the social ranking depends only on the *utility vectors* attached to the alternatives, not on the identity (names) of the alternatives themselves. If two pairwise comparisons involve the same utility vectors - just attached to different alternatives - the social ranking must be the same.

Remark (SN vs Arrow's Neutrality). Arrow's neutrality (Definition 8.1) states that permuting the labels of alternatives permutes the social ranking correspondingly. SN strengthens this: it replaces ordinal preference comparisons with identity of utility vectors. In the cardinally informed framework, SN implies both BI (take $x = z, y = w$) and PI (take $U = V, z = y, w = x$).

14.2 Welfarism Theorem

The Welfarism Theorem is a central result: under mild axioms, the SWFL collapses into a social ordering defined purely on utility space $\mathbb{R}^n$, removing all direct dependence on the underlying set of alternatives X .

Theorem 14.7. Welfarism Theorem

Let F be an SWFL with Universal Domain. The following three statements are equivalent:

(1) F satisfies **Binary Independence (BI)** and **Pareto Indifference (PI)**.

(2) F satisfies **Strong Neutrality (SN)**.

(3) There exists a rational (complete and transitive) preference relation R^* on $\mathbb{R}^n$ such that for all $x, y \in X$ and all $U \in D$,

$$xR_U y \iff U(x) R^* U(y). \quad (70)$$

The relation R^* is called the **social welfare ordering** on utility space.

Interpretation. Condition (3) means that the social ranking of any two alternatives depends *only* on the vectors of individual utilities they generate. The name of the alternative is irrelevant; all that matters is "who gets what." This is the **welfarist** principle at its core.

14.3 Proof of the Welfarism Theorem

Proof of (2) $\Rightarrow$ (1): SN implies BI and PI

SN $\Rightarrow$ BI. Take $z = x$ and $w = y$ in the definition of SN. If $U(x) = V(x)$ and $U(y) = V(y)$, then SN yields $xR_U y \iff xR_V y$, which is exactly BI.

SN $\Rightarrow$ PI. Take $V = U$, $z = y$, and $w = x$. If $U(x) = U(y)$, then SN gives $xR_U y \iff yR_U x$, i.e. $xI_U y$. This is exactly PI.

Proof of (1) $\Rightarrow$ (2): BI + PI $\Rightarrow$ SN

Goal. Assume $U(x) = V(z) = u$ and $U(y) = V(w) = v$. We must show $xR_U y \iff zR_V w$.

Setup (by Universal Domain). Since $D = \mathcal{U}$, we may freely construct auxiliary profiles $\bar{U}, \tilde{U}, \hat{U} \in D$ and an auxiliary alternative $\bar{x} \in X$ such that the utility assignments below hold. Here $\bar{x}$ carries the same utility vector u as x and z , serving as a "stepping stone" for PI.

	x	z	y	w	$\bar{x}$
U	u		v		
$\bar{U}$	u		v		u
$\tilde{U}$			v	v	u
$\hat{U}$		u		v	u
V		u		v	

Proof chain. The four steps below connect $xR_U y$ to $zR_V w$ via BI, PI, and transitivity of R :

$$\underbrace{xR_U y}_{\text{start}} \iff \underbrace{\bar{x}R_{\bar{U}} y}_{\text{Step 1}} \iff \underbrace{\bar{x}R_{\tilde{U}} w}_{\text{Step 2}} \iff \underbrace{zR_{\hat{U}} w}_{\text{Step 3}} \iff \underbrace{zR_V w}_{\text{Step 4 (end)}}. \quad (71)$$

Step 1 - from U to $\bar{U}$ (replace x by $\bar{x}$):

	Justification	Result
BI	$\bar{U}$ agrees with U on $\{x, y\}$	$xR_U y \iff xR_{\bar{U}} y$
PI	$\bar{U}(x) = \bar{U}(\bar{x}) = u$	$xI_{\bar{U}} \bar{x}$
Transitivity	combine the above	$xR_{\bar{U}} y \iff \bar{x}R_{\bar{U}} y$

Step 2 - from $\bar{U}$ to $\widetilde{U}$ (replace y by w):

	Justification	Result
BI	$\bar{U}, \widetilde{U}$ agree on $\{y, \bar{x}\}$	$\bar{x}R_{\bar{U}} y \iff \bar{x}R_{\widetilde{U}} y$
PI	$\widetilde{U}(y) = \widetilde{U}(w) = v$	$yI_{\widetilde{U}} w$
Transitivity	combine the above	$\bar{x}R_{\widetilde{U}} y \iff \bar{x}R_{\widetilde{U}} w$

Step 3 - from $\widetilde{U}$ to $\widehat{U}$ (replace $\bar{x}$ by z):

	Justification	Result
BI	$\widetilde{U}, \widehat{U}$ agree on $\{w, \bar{x}\}$	$\bar{x}R_{\widetilde{U}} w \iff \bar{x}R_{\widehat{U}} w$
PI	$\widehat{U}(\bar{x}) = \widehat{U}(z) = u$	$\bar{x}I_{\widehat{U}} z$
Transitivity	combine the above	$\bar{x}R_{\widehat{U}} w \iff zR_{\widehat{U}} w$

Step 4 - from $\widehat{U}$ to V :

	Justification	Result
BI	$\widehat{U}, V$ agree on $\{z, w\}$	$zR_{\widehat{U}} w \iff zR_V w$

Chaining Steps 1-4 yields $xR_U y \iff zR_V w$, i.e. **SN**. $\square$

Proof of (3) => (2): trivial

If (3) holds, then $xR_U y \iff U(x)R^*U(y)$. Under the premise $U(x) = V(z)$ and $U(y) = V(w)$, we obtain

$$xR_U y \iff U(x)R^*U(y) \iff V(z)R^*V(w) \iff zR_V w, \quad (72)$$

which is precisely SN.

Proof of (2) => (3): construction of R^*

Define R^* on utility space $\mathbb{R}^n$ as follows: for any $u, v \in \mathbb{R}^n$,

$$uR^*v \iff \exists x, y \in X, \exists U \in D \text{ such that } U(x) = u, U(y) = v, \text{ and } xR_U y. \quad (73)$$

We must verify that R^* is well-defined and rational.

Well-definedness. Suppose there exist two different profiles $U, V \in D$ and pairs of alternatives $(x, y), (z, w)$ such that $U(x) = u = V(z)$ and $U(y) = v = V(w)$. Strong Neutrality (SN) ensures

$$xR_U y \iff zR_V w, \quad (74)$$

so the definition of uR^*v is independent of the particular profile and alternatives chosen.

Completeness. For any $u, v \in \mathbb{R}^n$, by Universal Domain construct a profile U and alternatives x, y such that $U(x) = u, U(y) = v$. Since R_U is complete, either $xR_U y$ or $yR_U x$ (or both). Hence uR^*v or vR^*u (or both).

Transitivity. Let uR^*v and vR^*w . By definition, there exist profiles $U, V \in D$ and alternatives such that $xR_U y$ (with $U(x) = u, U(y) = v$) and $y'R_V z'$ (with $V(y') = v, V(z') = w$). By Universal Domain, construct a single profile W with three alternatives a, b, c such that $W(a) = u, W(b) = v, W(c) = w$. By SN (or equivalently BI+PI), the social rankings are preserved:

$$aR_W b \quad \text{and} \quad bR_W c. \quad (75)$$

Transitivity of R_W then implies $aR_W c$, hence uR^*w . So R^* is transitive.

Thus R^* is a rational preference on $\mathbb{R}^n$ and satisfies $xR_U y \iff U(x)R^*U(y)$ for all profiles. $\square$

15. From Social Welfare Ordering to Social Welfare Functions

Once Welfarism reduces the problem to a preference R^* on $\mathbb{R}^n$, the next natural question is: when can R^* be represented by a real-valued **social welfare function** $W : \mathbb{R}^n \rightarrow \mathbb{R}$?

Proposition 15.1. Continuity $\Rightarrow$ SWF Representation

If the social welfare ordering R^* on $\mathbb{R}^n$ satisfies **continuity** (i.e., for every $u \in \mathbb{R}^n$, the upper and lower contour sets $\{v \mid vR^*u\}$ and $\{v \mid uR^*v\}$ are closed), then by Debreu's representation theorem (cf. Proposition 3.6), there exists a continuous function $W : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$uR^*v \iff W(u) \geq W(v). \quad (76)$$

Combined with Welfarism, the SWFL takes the form

$$xR_U y \iff W(U(x)) \geq W(U(y)). \quad (77)$$

This is a **welfarist social welfare function**: society ranks alternatives by evaluating W at the utility vectors they generate.

15.1 Special Cases of Social Welfare Functions

Different normative principles correspond to different functional forms of W .

15.1.1 Generalized Utilitarianism

$$W(u) = \sum_{i=1}^n g_i(u_i), \quad g_i : \mathbb{R} \rightarrow \mathbb{R} \text{ (increasing transformation functions)}. \quad (78)$$

- **Standard (Benthamite) utilitarianism:** $g_i(u_i) = u_i$, i.e. $W(u) = \sum_{i=1}^n u_i$. Simple sum-ranking.
- **Concave** g_i (e.g. $g_i(u_i) = \sqrt{u_i}$ or $\log u_i$): the SWF exhibits inequality aversion - an extra unit of utility to a worse-off individual counts for more.
- The transformation g_i can be person-specific to incorporate differing needs or merit weights.

Connection to Separability. The generalized utilitarian form $W(u) = \sum_i g_i(u_i)$ is **additively separable** across individuals. This is a natural consequence if R^* satisfies strong separability over the n individual coordinates (cf. Proposition 5.5).

15.1.2 Maxmin (Rawlsian) Social Welfare Function

$$W(u) = \min_{i \in N} \{u_1, u_2, \dots, u_n\}. \quad (79)$$

- Society's welfare is measured by the utility of the **worst-off** individual.
- This is a Leontief-style social welfare function: improvements accrue to social welfare only insofar as they raise the floor.
- The indifference curves in utility space are L-shaped: society is indifferent between two distributions if and only if the minimum utility is the same.

Contrast with Utilitarianism. Utilitarianism permits compensating a rich individual's gain against a poor individual's loss as long as the sum increases. Maxmin rejects such trade-offs unless the poorest benefits.

15.1.3 Generalized Gini and Intermediate Forms

Between the extremes of utilitarian sum-ranking and Rawlsian maxmin lie families of inequality-averse SWFs:

- **CES (Constant Elasticity of Substitution):**

$$W(u) = \left(\sum_{i=1}^n u_i^{1-\varepsilon} \right)^{\frac{1}{1-\varepsilon}}, \quad \varepsilon \geq 0, \varepsilon \neq 1. \quad (80)$$

- $\varepsilon = 0$: Utilitarian.
- $\varepsilon \rightarrow \infty$: Maxmin.
- Intermediate ε : varying degrees of inequality aversion.

- **Generalized Gini:**

$$W(u) = \sum_{i=1}^n a_i u_{(i)}, \quad (81)$$

where $u_{(1)} \leq u_{(2)} \leq \dots \leq u_{(n)}$ are ordered utilities and $a_1 \geq a_2 \geq \dots \geq a_n$ are weights that give more importance to lower-ranked individuals.

Remark 15.2. Separability and the Welfarist Framework

The **separability** of R^* on $\mathbb{R}^n$ (cf. Sections 5.2-5.3) is an important structural condition:

- **Weak separability** of R^* across individuals means the ranking of u_i vs v_i (individual i 's utility) is independent of others' utility levels. This yields $W(u) = f(w_1(u_1), \dots, w_n(u_n))$.
- **Strong (additive) separability** further forces $W(u) = \sum_{i=1}^n w_i(u_i)$, the generalized utilitarian form.

Thus the additive-utilitarian structure is not an arbitrary assumption - it follows from reasonable separability axioms, paralleling the additively separable representation for consumer preferences (Proposition 5.5).

16. Information Invariance and Measurability of Utility

16.1 Review of the Welfarism Setup

Recall the SWFL framework (Section 14):

- $U_i : X \rightarrow \mathbb{R}$: individual utility function for $i \in N$.
- $D \subseteq \mathcal{U}$: domain of admissible utility profiles.
- $F : D \rightarrow \mathcal{R}$: social welfare functional mapping each profile $U = (U_1, \dots, U_n)$ to a social preference R_U .
- **Welfarism Theorem**: $\exists R^*$ on $\mathbb{R}^n$ such that $\forall u, v \in \mathbb{R}^n$,

$$u R^* v \iff x R_U y \text{ for some } x, y \in X, U \in \mathcal{U} \text{ with } U(x) = u, U(y) = v. \quad (82)$$

16.2 The Crucial Difference: Welfarism vs Arrow

The fundamental divergence between Arrow's framework (Sections 11-12) and the welfarist framework (Sections 14-15) lies in **information invariance** - what class of utility transformations leaves the social ranking unchanged.

Arrow's framework works with **ordinal** information only. Preferences are **ordinally comparable (OC)**: for any profile $V \in D$ and any vector of strictly increasing transformations $\phi = (\phi_1, \dots, \phi_n)$,

$$R_V = R_{\phi \circ V}, \quad \phi \circ V := (\phi_1 \circ V_1, \dots, \phi_n \circ V_n). \quad (83)$$

That is, applying any positive monotone transformation to individual utilities leaves the social ranking unchanged - only the ordinal ranking of alternatives matters.

Key observation. In the SWFL framework, BI (Binary Independence) + OC = Arrow's IIA. Under OC, the Welfarist Theorem collapses to an impossibility: **BI+OC implies dictatorship** (the Arrow-style result re-emerges).

Intuition. OC strips away all cardinal information, reducing the SWFL to an ordinal SWF. Once only ordinal rankings survive, Arrow's impossibility logic takes over.

16.3 Invariance Classes

An SWFL F is **invariant** under a class of admissible transformations Φ if for every profile $U \in D$ and every $\varphi \in \Phi$,

$$R_U = R_{\varphi \circ U}, \quad (84)$$

where $\varphi \circ U$ applies φ_i to individual i 's utility. The larger Φ is, the less information F can use; the smaller Φ , the less restricted F is.

Class	Admissible Transformations	Description
OC (Ordinal Comparability)	$\varphi_i(u_i)$ any strictly increasing	Only ordinal rankings preserved
CUC (Cardinal Unit Comparability)	$\varphi_i(u_i) = a_i + b u_i, a_i \in \mathbb{R}, b > 0$	Common unit; personal origins differ
TSM (Translation-Scale Measurability)	$\varphi_i(u_i) = a_i + u_i, a_i \in \mathbb{R}$	Only translation invariance
CFC (Cardinal Full Comparability)	$\varphi_i(u_i) = a + b u_i, a \in \mathbb{R}, b > 0$	Common unit and common origin
NM (Numerical Measurability)	$\varphi_i(u_i) = u_i$	Utility fully cardinal, absolutely measured

Nesting. $NM \subset CFC \subset CUC \subset OC$. The smaller Φ , the weaker the restriction on F - more information about the utility structure is available to the social planner.

16.4 Interpersonal Comparability

	Intra-person	Inter-person (level)	Inter-person (differences)
CUC	Yes	No	Yes (common b)
CFC	Yes	Yes (common a)	Yes (common b)

- **CUC** allows comparison of utility *differences* across individuals (same unit b), but not absolute utility *levels* (different personal origins a_i).
- **CFC** allows comparison of both levels and differences across individuals.

Under CUC and TSM, the ratio of utility differences is preserved:

$$\frac{\Delta u'_1}{\Delta u'_2} = \frac{\Delta u_1}{\Delta u_2}. \quad (85)$$

17. Axioms on the Utility Space $\mathbb{R}^n$

Once Welfarism collapses the problem to a social welfare ordering R on $\mathbb{R}^n$, we can impose axioms directly on utility vectors. Let R be a preference relation on $\mathbb{R}^n$.

Definition 17.1. Anonymity

For any $u \in \mathbb{R}^n$ and any permutation π of N ,

$$\pi \circ u \ I \ u. \quad (86)$$

Interpretation. Individual identities are irrelevant; only the distribution of utilities matters.

Definition 17.2. Strong Pareto

For any $u, v \in \mathbb{R}^n$,

$$u \geq v \text{ (componentwise) and } \exists j \in N \text{ s.t. } u_j > v_j \implies u P v. \quad (87)$$

Interpretation. If at least one person is strictly better off and no one is worse off, society strictly prefers the new distribution.

Definition 17.3. Weak Pareto

For any $u, v \in \mathbb{R}^n$,

$$u \gg v \text{ (} u_i > v_i \text{ for every } i \in N) \implies u P v. \quad (88)$$

Definition 17.4. Continuity

For any $u \in \mathbb{R}^n$, the upper contour set $\{v \in \mathbb{R}^n \mid v R u\}$ and lower contour set $\{v \in \mathbb{R}^n \mid u R v\}$ are **closed** in $\mathbb{R}^n$.

Definition 17.5. Separability

For any subset $M \subset N$, any $u_M, v_M \in \mathbb{R}^{|M|}$, and any $w_{-M}, w'_{-M} \in \mathbb{R}^{n-|M|}$,

$$(u_M, w_{-M}) R (v_M, w_{-M}) \iff (u_M, w'_{-M}) R (v_M, w'_{-M}). \quad (89)$$

Interpretation. The ranking of utilities within group M is independent of the utility levels of individuals outside M . This is the utility-space analogue of consumption separability (Section 5).

18. Characterization of Generalized Utilitarianism

Theorem 18.1. Debreu's Characterization

Let R be a social welfare ordering on $\mathbb{R}^n$. The following are equivalent:

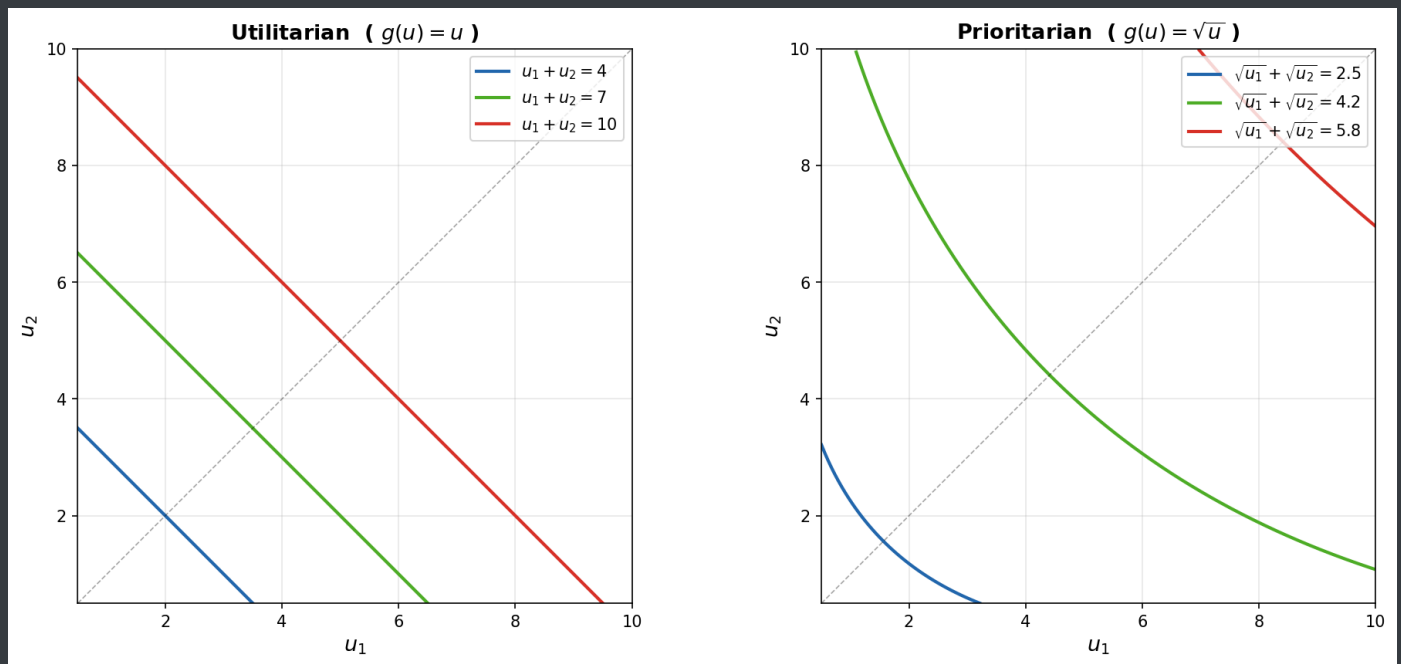
(i) R satisfies **Anonymity**, **Strong Pareto**, **Continuity**, and **Separability**.

(ii) R is **Generalized Utilitarian**: there exists a strictly increasing, continuous function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$uRv \iff \sum_{i \in N} g(u_i) \geq \sum_{i \in N} g(v_i). \quad (90)$$

Interpretation. The four axioms force social welfare to be an additively separable sum of transformed individual utilities. This parallels the consumer-theoretic result that strong separability yields additive utility (Proposition 5.5).

18.2 The Two-Person Illustration



For $n = 2$, the indifference curves of a generalized utilitarian SWF are given by

$$g(u_1) + g(u_2) = \text{constant}. \quad (91)$$

If g is linear, these are straight lines with slope -1 (standard utilitarianism). If g is strictly concave, indifference curves are convex to the origin - reflecting **inequality aversion**.

18.3 Concavity and Inequality Aversion

The shape of g encodes society's attitude toward inequality:

- g **linear** (e.g. $g(u_i) = u_i$): **Utilitarian** - society cares only about total utility; an extra unit to the rich is as valuable as to the poor.
- g **strictly concave** (e.g. $g(u_i) = \sqrt{u_i}$ or $\log u_i$): **Prioritarianism** - society gives greater weight to utility gains for the worse-off.

Connection to Section 15.1. The generalized utilitarian form $W(u) = \sum_i g(u_i)$ nests Benthamite utilitarianism ($g(u_i) = u_i$) as a special case. The concavity of g captures the same idea as the CES parameter ε in Section 15.1.3.

19. Equity Principles

19.1 Minimal Equity (ME)

Definition 19.1. R satisfies **Minimal Equity** if there exist $i, j \in N$ and $u, v \in \mathbb{R}^n$ such that

$$v_i < u_i < u_j < v_j, \quad u_k = v_k \text{ for all } k \neq i, j, \quad u R v. \quad (92)$$

Interpretation. Starting from an unequal distribution v , a Pigou-Dalton-style transfer from the richer individual j to the poorer individual i (producing u) is at least weakly socially preferred. ME is the weakest possible equity condition - it only requires that *some* equalizing transfer is viewed favorably.

19.2 Prioritarianism from Generalized Utilitarianism + ME

Proposition 19.2. If R is Generalized Utilitarian and satisfies Minimal Equity, then the transformation function g must be **concave**.

Interpretation. The combination of additive separability and even minimal equity concern forces the SWF to be inequality-averse. This is **Prioritarianism**: society's objective takes the form $\sum_i g(u_i)$ with g strictly increasing and concave, giving greater weight to the worse-off.

19.3 Incremental Equity (IE)

Definition 19.3. R satisfies **Incremental Equity** if for any $i, j \in N$, $u \in \mathbb{R}^n$, and $\delta \in \mathbb{R}$,

$$(u + \delta e_i) I (u + \delta e_j), \quad (93)$$

where e_i is the i -th unit vector.

Interpretation. Giving an incremental amount δ of utility to person i or to person j yields the same social welfare - society is indifferent about *who* receives an equal-sized gain.

Lemma 19.4. Anonymity + TSM $\implies$ IE

Proof. Take $u \in \mathbb{R}^n$ and $\delta \in \mathbb{R}$. By TSM (translation invariance),

$$(u + \delta e_i) R (u + \delta e_j) \iff \delta e_i R \delta e_j. \quad (94)$$

By Anonymity, $\delta e_i I \delta e_j$ (the two vectors are permutations of each other). Hence

$$(u + \delta e_i) I (u + \delta e_j). \quad \blacksquare \quad (95)$$

Theorem 19.5. Anonymity + TSM + Strong Pareto $\implies$ Utilitarianism

Under Universal Domain, if R satisfies **Anonymity**, **TSM** (Translation-Scale Measurability), and **Strong Pareto**, then R is **Utilitarian**:

$$uRv \iff \sum_{i \in N} u_i \geq \sum_{i \in N} v_i. \quad (96)$$

Proof. By Lemma 19.4, Anonymity + TSM $\Rightarrow$ IE. We show that under IE + Strong Pareto, every utility vector is indifferent to its **equalized mean**.

For any $u \in \mathbb{R}^n$, we construct a chain of IE transfers to redistribute utility until all coordinates equal $\frac{1}{n} \sum_i u_i$:

1. Transfer from person 1 to person n :

$$u \ I \ (u_1 - \delta_1, u_2, \dots, u_{n-1}, u_n + \delta_1) \quad \text{by IE, for any } \delta_1 \in \mathbb{R}. \quad (97)$$

Choose $\delta_1 = u_1 - \frac{1}{n} \sum_i u_i$, so the first coordinate becomes $\frac{1}{n} \sum_i u_i$, and the last coordinate becomes $u_n + u_1 - \frac{1}{n} \sum_i u_i$.

2. Transfer from person 2 to person n :

$$I \left(\frac{\sum u_i}{n}, \frac{\sum u_i}{n}, u_3, \dots, u_{n-1}, u_n + u_1 + u_2 - \frac{2 \sum u_i}{n} \right). \quad (98)$$

3. Continue iteratively through all coordinates. After n steps:

$$u \ I \ \left(\frac{\sum u_i}{n}, \frac{\sum u_i}{n}, \dots, \frac{\sum u_i}{n} \right). \quad (99)$$

Denote $\bar{u} = \frac{1}{n} \sum_i u_i$ and let $\bar{u} \cdot \mathbf{1}_n$ be the equalized vector. Then $u \ I \ \bar{u} \cdot \mathbf{1}_n$. Similarly, $v \ I \ \bar{v} \cdot \mathbf{1}_n$.

By Strong Pareto, $\bar{u} \cdot \mathbf{1}_n \ R \ \bar{v} \cdot \mathbf{1}_n \iff \bar{u} \geq \bar{v}$. Thus

$$uRv \iff \bar{u} \geq \bar{v} \iff \sum_i u_i \geq \sum_i v_i. \quad \blacksquare \quad (100)$$

Corollary 19.6. Anonymity + TSM + Strong Pareto $\implies$ Utilitarianism. This result shows that with translation-scale measurability (TSM) and a few mild axioms, the only admissible social welfare ordering is the sum-ranking.

20. Maximin and Leximin

20.1 Maximin (Rawlsian) Social Welfare Ordering

The **Maximin** rule evaluates distributions solely by the utility of the worst-off individual:

$$uRv \iff \min_{i \in N} u_i \geq \min_{i \in N} v_i. \quad (101)$$

- Society is indifferent between two distributions if and only if the minimum utility is the same.
- Improvements benefit social welfare only insofar as they raise the floor.
- Indifference curves in $\mathbb{R}^n$ are L-shaped: $uIv \iff \min_i u_i = \min_i v_i$.

Contrast with Utilitarianism. Under utilitarianism, society can trade off the poor's pain against the rich's gain provided the sum increases. Maximin rejects all such compensation unless the poorest benefits.

20.2 Leximin: Refining Maximin

Maximin ignores what happens above the floor - it cannot distinguish between $(0, 100, 100)$ and $(0, 0, 100)$. The **Leximin** ordering refines maximin by lexicographically comparing sorted utility vectors.

Definition 20.1 (Leximin). Let $u_{[1]} \leq u_{[2]} \leq \dots \leq u_{[n]}$ denote the components of u sorted in increasing order. Then

$$uPv \iff \exists T \in \{0, \dots, n-1\} \text{ such that } u_{[t]} = v_{[t]} \text{ for } t = 1, \dots, T \text{ and } u_{[T+1]} > v_{[T+1]}. \quad (102)$$

Equivalently: compare the smallest utilities first; if tied, compare the second-smallest; and so on. Indifference uIv holds $\iff u_{[t]} = v_{[t]}$ for all $t = 1, \dots, n$.

Observation. $uIv \iff v = \pi \circ u$ for some permutation π - leximin indifference coincides with Anonymity.

Example 20.2. Leximin Comparison

Let $u = (3, 0, 1, 4)$ and $v = (0, 2, 1, 2)$. Sorted:

$$\hat{u} = (0, 1, 3, 4), \quad \hat{v} = (0, 1, 2, 2). \quad (103)$$

Compare lexicographically:

- $t = 1: \hat{u}_{[1]} = 0 = \hat{v}_{[1]}$ (tie)
- $t = 2: \hat{u}_{[2]} = 1 = \hat{v}_{[2]}$ (tie)
- $t = 3: \hat{u}_{[3]} = 3 > \hat{v}_{[3]} = 2$

Hence uPv .

Remark 20.3. Leximin and Separability

Leximin violates **separability** (Definition 17.5): the ranking of two individuals' utilities can depend on the utility levels of a third individual. This is intuitive - leximin cares about the *ordering* of all individuals, so adding a very poor person can affect the social ranking of two richer individuals even if their own utilities are unchanged.

20.4 Review (Lecture 7, 6/25): Maximin and Leximin

For any $u, v \in \mathbb{R}^n$:

Maximin.

$$uRv \iff \min_{i \in N} u_i \geq \min_{i \in N} v_i. \quad (104)$$

Leximin. Let $u_{[1]} \leq u_{[2]} \leq \dots \leq u_{[n]}$ denote the components of u sorted in increasing order. Then

$$uPv \iff \exists T \in \{0, \dots, n-1\} \text{ such that } u_{[t]} = v_{[t]} \text{ for } t = 1, \dots, T \text{ and } u_{[T+1]} > v_{[T+1]}. \quad (105)$$

(See Definition 20.1 for the equivalent formulation.)

20.5 Geometric Illustration in $\mathbb{R}^2$

For $n = 2$, both Maximin and Leximin yield **L-shaped indifference curves** through any utility vector $u = (u_1, u_2)$. The vertex of the L lies on the 45° line ($u_1 = u_2$) at $\min\{u_1, u_2\}$.

Figure (Maximin vs. Leximin: indifference curves and strictly preferred/dispreferred sets).

7th 6/25

Review.

• maximin: for any $u, v \in \mathbb{R}^n$.

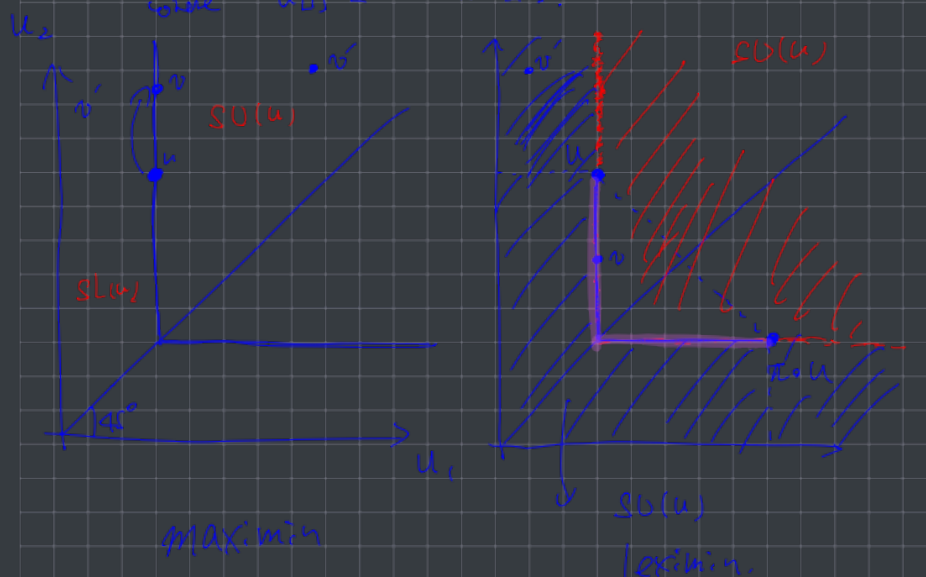
$$u R v \Leftrightarrow \min_{i \in n} u_i \geq \min_{i \in n} v_i.$$

• leximin: for any $u, v \in \mathbb{R}^n$

$$u P v \Leftrightarrow u[i] = v[i] \quad \forall i = 0 \dots L$$

$$u[L+1] > v[L+1]$$

$$\text{where } u[i] \leq \dots \leq u[n].$$



Maximin geometry

For any $u \in \mathbb{R}^2$, define:

- $SU(u)$: the **strictly upper** set - all v with $\min_i v_i > \min_i u_i$.
- $SL(u)$: the **strictly lower** set - all v with $\min_i v_i < \min_i u_i$.

Under Maximin, $u I v$ whenever $\min_i u_i = \min_i v_i$. Points on the same L-shaped boundary - e.g. moving from u vertically upward to v while keeping the same minimum - are **indifferent**. Only moves that strictly raise the floor lie in $SU(u)$.

Leximin geometry

Leximin refines Maximin. With $u_{[1]} \leq u_{[2]}$, the ordering compares sorted coordinates lexicographically:

- $SU(u)$: the shaded region **inside** the L (strictly preferred) - includes points where the minimum is higher, or the minimum is tied but the second-smallest utility is strictly higher.
- $SD(u)$: the shaded region **outside** the L (strictly dispreferred).

Thus, for $u = (a, a)$ with $a = \min\{u_1, u_2\}$, a point v on the vertical arm above u (same minimum, higher maximum) satisfies vPu under Leximin - but uIv under Maximin. This illustrates Leximin's refinement of the floor rule.

21. Hammond Equity

Definition 21.1 (Hammond Equity)

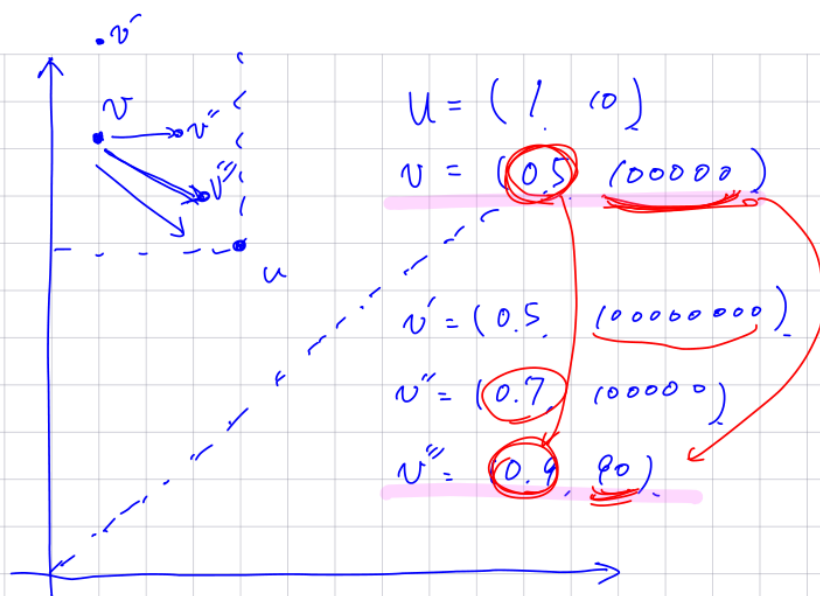
R satisfies **Hammond Equity (HE)** if for any $u, v \in \mathbb{R}^n$ such that

$$v_i < u_i \leq u_j < v_j, \quad u_k = v_k \text{ for all } k \neq i, j, \quad (106)$$

we have uRv .

Interpretation. Starting from an unequal distribution v , a transfer from the richer individual j to the poorer individual i that reduces inequality (while keeping i no better off than j) is weakly socially preferred. Hammond Equity is stronger than **Minimal Equity** (Definition 19.1), which only requires that *some* equalizing transfer is favored - it does not require the pattern $v_i < u_i \leq u_j < v_j$ for every favored transfer.

Figure (Hammond Equity: equalizing transfer in $\mathbb{R}^2$).



Hammond Equity.

For any $u, v \in \mathbb{R}^n$ s.t.

$$v_i < u_i \leq u_j < v_j$$

$$u_k = v_k \quad \text{for any } k \neq i, j.$$

then, $u R v$

Example 21.2. Hammond Equity Transfers

Let $u = (1, 10)$. Consider increasingly unequal distributions and their progressive equalization:

$$v = (0.5, 1,000,000), \quad v'' = (0.7, 100,000), \quad v''' = (0.9, 10). \quad (107)$$

Each step moves the distribution closer to $u = (1, 10)$ by reducing the gap between the two individuals. By Hammond Equity, Pareto comparisons where needed, and transitivity, $uRv'''Rv''Rv$: society prefers the more equal distribution.

22. Characterization of Leximin and Maximin

Theorem 22.1. Leximin Characterization

Let R be a social welfare ordering on $\mathbb{R}^n$. The following are equivalent:

(i) R satisfies **Anonymity**, **Hammond Equity**, and **Strong Pareto**.

(ii) R is **Leximin** (Definition 20.1).

Proof sketch ($\Rightarrow$). By Anonymity, we may assume utilities are sorted: $u_1 \leq \dots \leq u_n$. Take any $u, v \in \mathbb{R}^n$ with uPv under Leximin - i.e., there exists T such that $u_{[t]} = v_{[t]}$ for $t = 1, \dots, T$ and $u_{[T+1]} > v_{[T+1]}$.

Construct a fully equalized reference vector

$$v^* = (\varepsilon_1, \varepsilon_1 + \varepsilon_2, \varepsilon_1 + \varepsilon_2 + \varepsilon_3, \dots, \sum_{i=1}^n \varepsilon_i), \quad (108)$$

where each $\varepsilon_i > 0$ is chosen small enough.

Using **Hammond Equity**, we transform v step by step toward v^* by compressing inequality between adjacent coordinates:

$$\begin{aligned} v &= (v_1, v_2, v_3, \dots, v_n) \\ R(v_1 \wedge v_2, v_1 \vee v_2, v_3, \dots, v_n) \\ R(\varepsilon_1, \varepsilon_1 + \varepsilon_2, \varepsilon_1 + \varepsilon_2, v_4, \dots, v_n) \\ R \dots R v^*. \end{aligned} \quad (109)$$

Each step applies Hammond Equity to a pair (i, j) where a transfer from the richer to the poorer individual is socially favored.

By **Strong Pareto**, for $\varepsilon_i > 0$ small enough, uPv^* . By transitivity, uPv .

Generalization. If $u_{[t]} = v_{[t]}$ for $t = 1, \dots, T$ and $u_{[T+1]} > v_{[T+1]}$, then uPv . ■

Theorem 22.2. Maximin Characterization

Let R be a social welfare ordering on $\mathbb{R}^n$. The following are equivalent:

(i) R satisfies **Hammond Equity**, **Weak Pareto**, and **Continuity**.

(ii) R is **Maximin** (Section 20.1).

Remark. Anonymity can be dropped from the axiom list - Hammond Equity together with Weak Pareto and Continuity already forces the Maximin rule.

Remark 22.3 (Incompatibility). Under Hammond Equity, **Strong Pareto** and **Continuity** are incompatible. This explains why the Leximin characterization uses Strong Pareto while the Maximin characterization uses only Weak Pareto plus Continuity.

23. Population Ethics: Variable Populations

So far we have compared utility profiles for a **fixed** population $N = \{1, \dots, n\}$. **Population ethics** extends the framework to **variable populations**: we may compare states in which different individuals exist.

23.1 Setup

- $\mathbb{N} = \{1, 2, 3, \dots\}$: the set of **potential individuals**.
- $N \subseteq \mathbb{N}$: a **finite** subset (a population).
- $\mathcal{U}^N = \mathbb{R}^{|N|}$: utility profiles of individuals in N .
- $\mathcal{U} = \bigcup_{\substack{N \subseteq \mathbb{N} \\ N \text{ finite}}} \mathcal{U}^N$: the **domain** of all utility profiles over all finite populations.

Example. $N = \{1, 2, \dots, 100\}$ and $M = \{3, 4, \dots, 10000\}$ are two populations. In general $N \cap M$ need not be empty, and $|N| \neq |M|$.

A **social welfare ordering (SWO)** R on $\mathcal{U}$ ranks utility profiles across potentially different populations:

$$u_N R u_M, \tag{110}$$

where $u_N \in \mathcal{U}^N$ and $u_M \in \mathcal{U}^M$. This is our objective.

23.2 Extension of Criteria

Generalized Utilitarianism (variable population). For strictly increasing $g : \mathbb{R} \rightarrow \mathbb{R}$,

$$u_N R^{GU} v_M \iff \sum_{i \in N} g(u_i) \geq \sum_{j \in M} g(v_j). \tag{111}$$

Remark. By suitably modifying the axioms from the fixed-population framework, one can obtain similar axiomatic foundations for variable-population SWOs.

24. The Repugnant Conclusion

24.1 Utilitarianism and Population Size

Under **Utilitarianism** in the variable-population setting,

$$u_N R^U v_M \iff \sum_{i \in N} u_i \geq \sum_{j \in M} v_j. \quad (112)$$

Example 24.1. Let $u_N = (100, 100, 100)$ with $|N| = 3$ and $v_M = (0.3, 0.3, \dots, 0.3)$ with $|M| = 10,000$. Then

$$\sum_{i \in N} u_i = 300 < 0.3 \times 10,000 = 3000 = \sum_{j \in M} v_j, \quad (113)$$

so $v_M P^U u_N$. A large population living at a barely positive utility level is strictly preferred to a small population living very well - simply because the total sum is larger.

Definition 24.2 (Repugnant Conclusion)

R induces the **Repugnant Conclusion** if for any $r > 0$, any finite population $N \subseteq \mathbb{N}$, and any $\epsilon \in (0, r)$, there exists a population $M \subseteq \mathbb{N}$ such that

$$\underbrace{(\epsilon, \dots, \epsilon)}_{|M|} P \underbrace{(r, \dots, r)}_{|N|}. \quad (114)$$

Observation. Utilitarianism induces the Repugnant Conclusion: by adding enough people at utility $\epsilon > 0$, the total sum eventually exceeds any fixed total for a smaller, better-off population.

25. Average and Critical-Level Utilitarianism

Two common modifications of utilitarianism attempt to avoid the Repugnant Conclusion.

25.1 Average Utilitarianism (AU)

$$u_N R^{AU} v_M \iff \frac{\sum_{i \in N} u_i}{|N|} \geq \frac{\sum_{j \in M} v_j}{|M|}. \quad (115)$$

Average Utilitarianism and Critical-Level Utilitarianism (with $\alpha > 0$) **can avoid** the Repugnant Conclusion.

25.2 Critical-Level Utilitarianism (CLU)

For a **critical level** $\alpha > 0$,

$$u_N R^{CLU} v_M \iff \sum_{i \in N} (u_i - \alpha) \geq \sum_{j \in M} (v_j - \alpha). \quad (116)$$

Each person's utility is counted only in excess of the critical level α . Adding a person with utility below α lowers the social welfare sum.

25.3 Criticism: Violation of Mere Addition

Both AU and CLU fail a basic population-ethics principle.

Example 25.3. Let $u_N = (100, 100, 100)$ and $v_M = (100, 100, 100, 0.1)$. Then

$$\frac{300}{3} = 100 > \frac{300.1}{4} \approx 75.03, \quad (117)$$

so $u_N P^{AU} v_M$. Adding a person with utility $0.1 > 0$ makes society strictly worse off under Average Utilitarianism. The same holds for CLU when $\alpha > 0.1$.

26. Mere Addition and the Sadistic Conclusion

Definition 26.1 (Mere Addition Principle)

R satisfies the **Mere Addition Principle (MAP)** if for any $u_N \in \mathcal{U}$ and any $v_k > 0$ (a person k not in N with strictly positive utility),

$$(u_N, v_k) R u_N. \quad (118)$$

Adding a person with strictly positive utility should not make society worse off.

Observation. Average Utilitarianism and Critical-Level Utilitarianism (with $\alpha > 0$) do **not** satisfy the Mere Addition Principle (Example 25.3).

Definition 26.2 (Sadistic Conclusion)

R induces the **Sadistic Conclusion** if there exist populations $N, M, L \subseteq \mathbb{N}$ with $L \cap M = \emptyset$ and $L \cap N = \emptyset$, utility profiles $u_M \in \mathbb{R}_{++}^{|M|}$ (all strictly positive), $u_N \in \mathbb{R}_{--}^{|N|}$ (all strictly negative), and $w_L \in \mathbb{R}^{|L|}$, such that

$$(u_N, w_L) P (u_M, w_L). \quad (119)$$

Society strictly prefers a state in which a group N of individuals with **negative** utility exists over a state in which a group M of individuals with **positive** utility exists - holding fixed the welfare of the background population w_L .

Observation. Critical-Level Utilitarianism can induce the Sadistic Conclusion when $\alpha > 0$: if adding a person with utility below α is bad, then comparisons can be constructed in which adding negative-utility lives is ranked above adding positive but sub-critical lives.

26.3 The Dilemma

Population ethics faces a fundamental trade-off:

Criterion	Utilitarianism	AU / CLU ($\alpha > 0$)
Repugnant Conclusion	Yes	No
Mere Addition Principle	Yes	No
Sadistic Conclusion	No	Yes (CLU)

There is no free lunch: avoiding the **Repugnant Conclusion** (via AU or CLU) comes at the cost of violating **Mere Addition**; Critical-Level Utilitarianism additionally risks the **Sadistic Conclusion**. Some trade-off among these principles is unavoidable.

Chapter 5. Mechanism Design

27. Mechanism Design and Strategy-Proof Social Choice

This section follows [new.pdf](#) , pp. 35-41.

27.1 Basic Setup

Let:

- A be the set of alternatives.
- $N = \{1, \dots, n\}$ be the set of individuals.
- $\succ_i \in \mathcal{R}$ be individual i 's rational preference.
- $D \subseteq \mathcal{R}^n$ be the domain of preference profiles.
- $f : D \rightarrow A$ be a **social choice function**.

The central question is how to construct an SCF that is robust against strategic reports.

27.2 Strategy-Proofness

The SCF f is **strategy-proof** if for every individual $i \in N$, every true profile $(\succ_i, \succ_{-i}) \in D$, and every possible report $\succ'_i$ with $(\succ'_i, \succ_{-i}) \in D$,

$$f(\succ_i, \succ_{-i}) \succeq_i f(\succ'_i, \succ_{-i}). \tag{120}$$

Truthful reporting weakly dominates every unilateral misreport.

27.3 Pareto Efficiency

The SCF f is **Pareto efficient** if for every profile $\succ \in D$, there is no $y \in A$ with $y \neq f(\succ)$ such that

$$y \succeq_i f(\succ) \text{ for all } i \in N, \quad y \succ_j f(\succ) \text{ for some } j \in N. \tag{121}$$

Thus the chosen alternative is never Pareto dominated by another alternative.

27.4 Gibbard–Satterthwaite Theorem

Suppose $|A| \geq 3$. An SCF

$$f : \mathcal{P}^n \rightarrow A \quad (122)$$

defined on the full domain of strict preferences satisfies **strategy-proofness** and **Pareto efficiency** if and only if f is **dictatorial**.

That is, there exists $i^* \in N$ such that for every profile $\succ \in \mathcal{P}^n$,

$$f(\succ) \succeq_{i^*} y \quad \text{for every } y \neq f(\succ). \quad (123)$$

27.5 Lower Contour Sets and Maskin Monotonicity

For any preference $\succ_i$ and any $a \in A$, define the lower contour set of a under $\succ_i$ by

$$L_i(\succ_i, a) = \{b \in A \mid a \succeq_i b\}. \quad (124)$$

A preference $\succ'_i$ is a **Maskin-monotonic transformation** of $\succ_i$ at a if

$$L_i(\succ_i, a) \subseteq L_i(\succ'_i, a). \quad (125)$$

Equivalently, a 's rank weakly moves upward: every alternative that was below a remains below a after the transformation.

The SCF f is **Maskin monotonic** if for every $\succ, \succ' \in D$, whenever $\succ'_i$ is a Maskin-monotonic transformation of $\succ_i$ at $f(\succ)$ for every $i \in N$,

$$f(\succ') = f(\succ). \quad (126)$$

Proposition 27.1. Strategy–Proofness and Maskin Monotonicity

For $f : \mathcal{P}^n \rightarrow A$,

$$f \text{ is strategy-proof} \iff f \text{ is Maskin monotonic}. \quad (127)$$

27.6 Proof Tables for the Dictatorship Argument

Take any two alternatives $a, b \in A$. The proof uses a sequence of preference profiles. Entries in each column are ordered from top to bottom; vertical dots indicate alternatives whose exact locations are not needed.

Step 1: Move b Up One Individual at a Time

Initial unanimous profile:

Profile (1)	Voters $1, \dots, n$
Top	a
	$\vdots$
Bottom	b

By Pareto efficiency,

$$f((1)) = a. \quad (128)$$

Final unanimous profile after moving b above a for everyone:

Profile (2)	Voters $1, \dots, n$
Top	b
Second	a
	$\vdots$

By Pareto efficiency,

$$f((2)) = b. \quad (129)$$

Therefore there exists a pivotal individual i^* such that the outcome changes from a to b when i^* moves b above a .

Before i^* moves b above a :

Profile (2)'	$1, \dots, i^* - 1$	i^*	$i^* + 1, \dots, n$
Higher	b	a	a
Lower	a	b	$\vdots$
Bottom	$\vdots$	$\vdots$	b

$$f((2)') = a. \quad (130)$$

After i^* moves b above a :

Profile (3)	$1, \dots, i^* - 1$	i^*	$i^* + 1, \dots, n$
Higher	b	b	a
Lower	a	a	$\vdots$
Bottom	$\vdots$	$\vdots$	b

$$f((3)) = b. \quad (131)$$

The proof then shows that this pivotal individual i^* is a dictator.

Step 2: Decrease the Rank of a

The note next lowers a while keeping the decisive comparison around b .

Profile (4)	$1, \dots, i^* - 1$	i^*	$i^* + 1, \dots, n$
Higher	b	b	$\vdots$
	$\vdots$	a	a
Lower	$\vdots$	$\vdots$	b
Bottom	a	$\vdots$	$\vdots$

$$f((4)) = b. \tag{132}$$

The next table pushes a further down while preserving the relevant monotonicity relation at b .

Profile (5)	$1, \dots, i^* - 1$	i^*	$i^* + 1, \dots, n$
Higher	b	a	$\vdots$
	$\vdots$	$\vdots$	a
Lower	$\vdots$	$\vdots$	b
Bottom	a	$\vdots$	$\vdots$

By Maskin monotonicity at b ,

$$f((5)) = b. \tag{133}$$

Step 3: Introduce a Third Alternative c

Take any $c \neq a, b$. The next profile places c above b for the relevant voters.

Profile (6)	$1, \dots, i^* - 1$	i^*	$i^* + 1, \dots, n$
Higher	$\vdots$	a	$\vdots$
	$\vdots$	c	$\vdots$
	c	b	c
Lower	b	$\vdots$	a
Bottom	a	$\vdots$	b

By Pareto efficiency, b cannot be chosen whenever c Pareto dominates b in this constructed profile. Combining this with the monotonicity transformation from the previous profile forces the outcome away from b in the direction needed to identify i^* as decisive.

The next profile keeps the same pivotal structure while shifting the relative position of b .

Profile (7)	$1, \dots, i^* - 1$	i^*	$i^* + 1, \dots, n$
Higher	$\vdots$	a	$\vdots$
	$\vdots$	c	$\vdots$
	c	b	a
Lower	b	$\vdots$	c
Bottom	a	$\vdots$	b

Pareto efficiency and monotonicity imply

$$f((7)) = a. \quad (134)$$

Finally, after the remaining irrelevant ranks are adjusted, the proof obtains a profile of the following form:

Profile (8)	$1, \dots, i^* - 1$	i^*	$i^* + 1, \dots, n$
Higher	$\vdots$	a	$\vdots$
	$\vdots$	$\vdots$	$\vdots$
Bottom	a	$\vdots$	a

$$f((8)) = a. \quad (135)$$

Since $a, b \in A$ were arbitrary, i^* can force the chosen alternative to follow his top-ranked alternative against every rival. Hence i^* is a dictator.

27.7 Restricted Domains and Single-Peaked Preferences

Once the full-domain assumption is dropped, positive results may be possible. One important domain restriction is **single-peakedness**.

A preference over a linearly ordered set of alternatives is single-peaked if it has one peak and becomes worse as alternatives move away from that peak on either side.

Let

$$n = 2m + 1, \quad m \geq 1. \quad (136)$$

An SCF

$$f : SP^n \rightarrow A \quad (137)$$

is called a **generalized median voter rule** if there exist fixed alternatives

$$\alpha_1, \dots, \alpha_{n-1} \in A \quad (138)$$

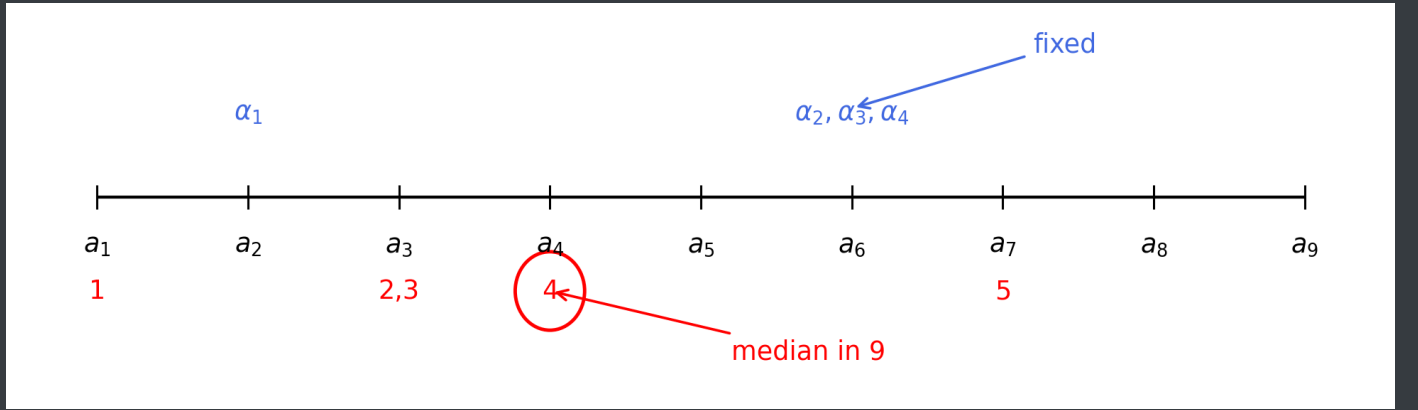
such that for every single-peaked profile $\succ \in SP^n$,

$$f(\succ) = \text{med}\{\tau(\succ_1), \tau(\succ_2), \dots, \tau(\succ_n), \alpha_1, \dots, \alpha_{n-1}\}, \quad (139)$$

where $\tau(\succ_i)$ is the peak of individual i 's preference.

The note's line diagram fixes alternatives

$$a_1 < a_2 < a_3 < a_4 < a_5 < a_6 < a_7 < a_8 < a_9 \quad (140)$$



and places the fixed alternatives at

$$\alpha_1 = a_2, \quad \alpha_2 = \alpha_3 = \alpha_4 = a_6, \quad (141)$$

with the median in a set of nine entries located at the fifth order statistic.

Theorem 27.2. Characterization on the Single-Peaked Domain

For $f : SP^n \rightarrow A$, the standard Moulin characterization says that

$$f \text{ is strategy-proof and peaks-only} \iff f \text{ is a generalized median voter rule.} \quad (142)$$

Adding anonymity is consistent with generalized median rules because the fixed phantom peaks do not depend on voter identities. Adding Pareto efficiency imposes boundary conditions on the phantom peaks; in the symmetric odd-voter case, this yields the ordinary median voter rule as the canonical example.